

Wazuh SIEM Installation and Nmap Detection

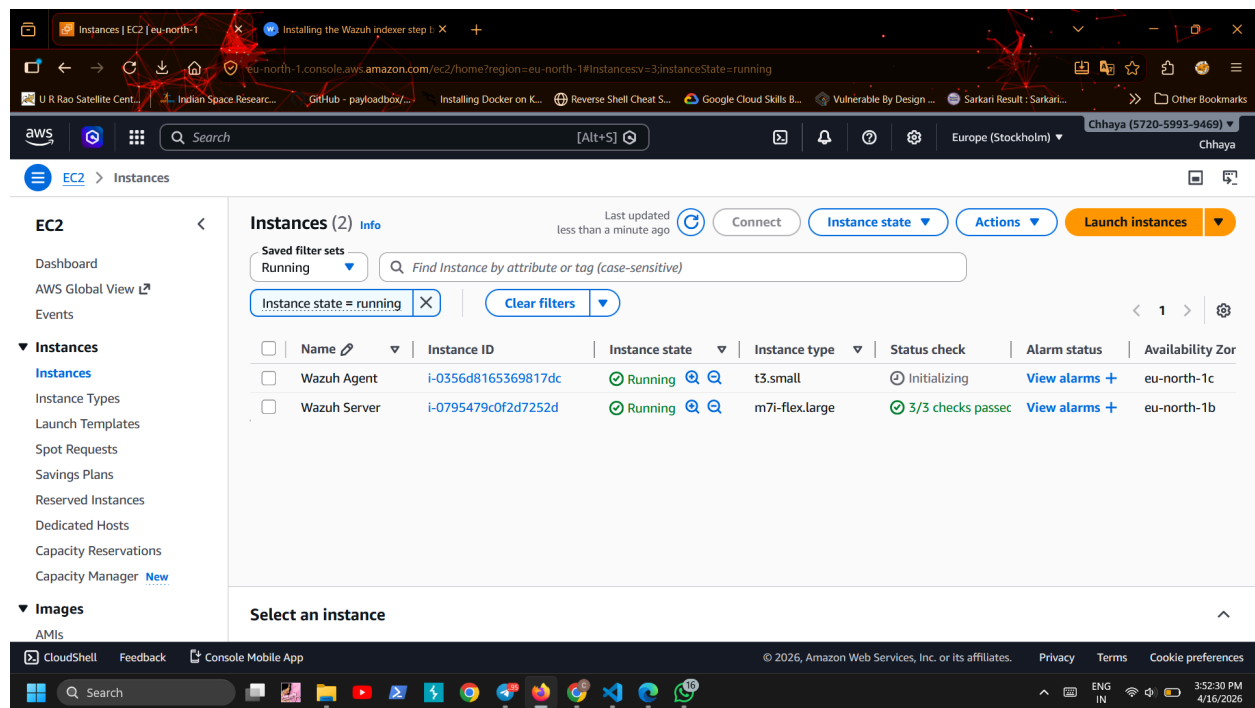
1. Objective

The objective of this task was to install Wazuh SIEM on a local machine (Wazuh Server) and connect another virtual machine (Wazuh Agent) to it. After completing the setup, an Nmap scan was performed on the agent machine to check whether Wazuh detects and generates alerts for such activity.

2. Lab Setup

For this lab, I used two machines:

- Wazuh Server – where Wazuh SIEM is installed
- Wazuh Agent – separate VM connected to the server



3. Wazuh Installation on Wazuh Server

I followed the official Wazuh documentation to install the indexer component. The process was divided into three main stages.

3.1 Certificate Creation

In the first step, I created SSL certificates required for secure communication between Wazuh components.

- Downloaded the certificate generation script and configuration file
- Edited the configuration file to add node names and IP addresses for:
 - Wazuh indexer
 - Wazuh server
 - Wazuh dashboard
- Ran the script to generate certificates
- Compressed the generated certificates into a single file
- Copied the certificate file to the required nodes

```

Get:44 http://security.ubuntu.com/ubuntu noble-security/main amd64 Components [21.5 kB]
Get:45 http://security.ubuntu.com/ubuntu noble-security/main amd64 c-n-f Metadata [10.8 kB]
Get:46 http://security.ubuntu.com/ubuntu noble-security/universe amd64 Packages [1169 kB]
Get:47 http://security.ubuntu.com/ubuntu noble-security/universe Translation-en [225 kB]
92% [Waiting for headers]
92% [Waiting for headers]
92% [Waiting for headers]
Get:48 http://security.ubuntu.com/ubuntu noble-security/universe amd64 Components [74.1 kB]
Get:49 http://security.ubuntu.com/ubuntu noble-security/universe amd64 c-n-f Metadata [22.9 kB]
Get:50 http://security.ubuntu.com/ubuntu noble-security/restricted amd64 Packages [2810 kB]
Get:51 http://security.ubuntu.com/ubuntu noble-security/restricted Translation-en [654 kB]
Get:52 http://security.ubuntu.com/ubuntu noble-security/restricted amd64 Components [212 B]
Get:53 http://security.ubuntu.com/ubuntu noble-security/multiverse amd64 Packages [28.8 kB]
Get:54 http://security.ubuntu.com/ubuntu noble-security/multiverse Translation-en [7204 B]
Get:55 http://security.ubuntu.com/ubuntu noble-security/multiverse amd64 Components [208 B]
Get:56 http://security.ubuntu.com/ubuntu noble-security/multiverse amd64 c-n-f Metadata [396 B]
Fetched 41.5 MB in 2min 31s (276 kB/s)
Reading package lists... Done
ubuntu@ip-172-31-34-18:~$ curl -sO https://packages.wazuh.com/4.14/wazuh-certs-tool.sh
curl -sO https://packages.wazuh.com/4.14/config.yml
ubuntu@ip-172-31-34-18:~$ nano config.yml
ubuntu@ip-172-31-34-18:~$ nano config.yml
ubuntu@ip-172-31-34-18:~$
  
```

i-0795479c0f2d7252d (Wazuh Server)

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GNU nano 7.2 config.yml

```
# Wazuh indexer nodes
indexer:
  - name: node-1
    ip: "172.31.34.18"
  # name: node-2
  # ip: "<indexer-node-ip>"
  # name: node-3
  # ip: "<indexer-node-ip>"

# Wazuh server nodes
# If there is more than one Wazuh server
# node, each one must have a node_type
server:
  - name: wazuh-1
    ip: "172.31.34.18"
  # node_type: master
  # name: wazuh-2
  # ip: "<wazuh-manager-ip>"
```

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GNU nano 7.2 config.yml

```
# Wazuh server nodes
# If there is more than one Wazuh server
# node, each one must have a node_type
server:
  - name: wazuh-1
    ip: "172.31.34.18"
  # node_type: master
  # name: wazuh-2
  # ip: "<wazuh-manager-ip>"
  # node_type: worker
  # name: wazuh-3
  # ip: "<wazuh-manager-ip>"
  # node_type: worker

# Wazuh dashboard nodes
dashboard:
  - name: dashboard
    ip: "172.31.34.18"
```

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```
Get:50 http://security.ubuntu.com/ubuntu noble-security/restricted amd64 Packages [2810 kB]
Get:51 http://security.ubuntu.com/ubuntu noble-security/restricted Translation-en [654 kB]
Get:52 http://security.ubuntu.com/ubuntu noble-security/restricted amd64 Components [212 B]
Get:53 http://security.ubuntu.com/ubuntu noble-security/multiverse amd64 Packages [28.8 kB]
Get:54 http://security.ubuntu.com/ubuntu noble-security/multiverse Translation-en [7204 B]
Get:55 http://security.ubuntu.com/ubuntu noble-security/multiverse amd64 Components [208 B]
Get:56 http://security.ubuntu.com/ubuntu noble-security/multiverse amd64 c-n-f Metadata [396 B]
Fetched 41.5 MB in 2min 31s (276 kB/s)
Reading package lists... Done
ubuntu@ip-172-31-34-18:~$ curl -sO https://packages.wazuh.com/4.14/wazuh-certs-tool.sh
curl -sO https://packages.wazuh.com/4.14/config.yml
ubuntu@ip-172-31-34-18:~$ nano config.yml
ubuntu@ip-172-31-34-18:~$ nano config.yml
ubuntu@ip-172-31-34-18:~$ nano config.yml
ubuntu@ip-172-31-34-18:~$ bash ./wazuh-certs-tool.sh -A
16/04/2026 10:46:05 INFO: Verbose logging redirected to /home/ubuntu/wazuh-certificates-tool.log
16/04/2026 10:46:05 INFO: Generating the root certificate.
16/04/2026 10:46:06 INFO: Generating Admin certificates.
16/04/2026 10:46:06 INFO: Admin certificates created.
16/04/2026 10:46:06 INFO: Generating Wazuh indexer certificates.
16/04/2026 10:46:06 INFO: Wazuh indexer certificates created.
16/04/2026 10:46:06 INFO: Generating Filebeat certificates.
16/04/2026 10:46:06 INFO: Wazuh Filebeat certificates created.
```

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```
16/04/2026 10:46:05 INFO: Generating the root certificate.
16/04/2026 10:46:06 INFO: Generating Admin certificates.
16/04/2026 10:46:06 INFO: Admin certificates created.
16/04/2026 10:46:06 INFO: Generating Wazuh indexer certificates.
16/04/2026 10:46:06 INFO: Wazuh indexer certificates created.
16/04/2026 10:46:06 INFO: Generating Filebeat certificates.
16/04/2026 10:46:06 INFO: Wazuh Filebeat certificates created.
16/04/2026 10:46:06 INFO: Generating Wazuh dashboard certificates.
16/04/2026 10:46:06 INFO: Wazuh dashboard certificates created.
ubuntu@ip-172-31-34-18:~$ tar -cvf ./wazuh-certificates.tar -C ./wazuh-certificates/ .
rm -rf ./wazuh-certificates
./
./node-1.pem
./dashboard-key.pem
./admin.pem
./admin-key.pem
./dashboard.pem
./root-ca.key
./wazuh-1.pem
./node-1-key.pem
./root-ca.pem
./wazuh-1-key.pem
ubuntu@ip-172-31-34-18:~$
```

i-0795479c0f2d7252d (Wazuh Server)

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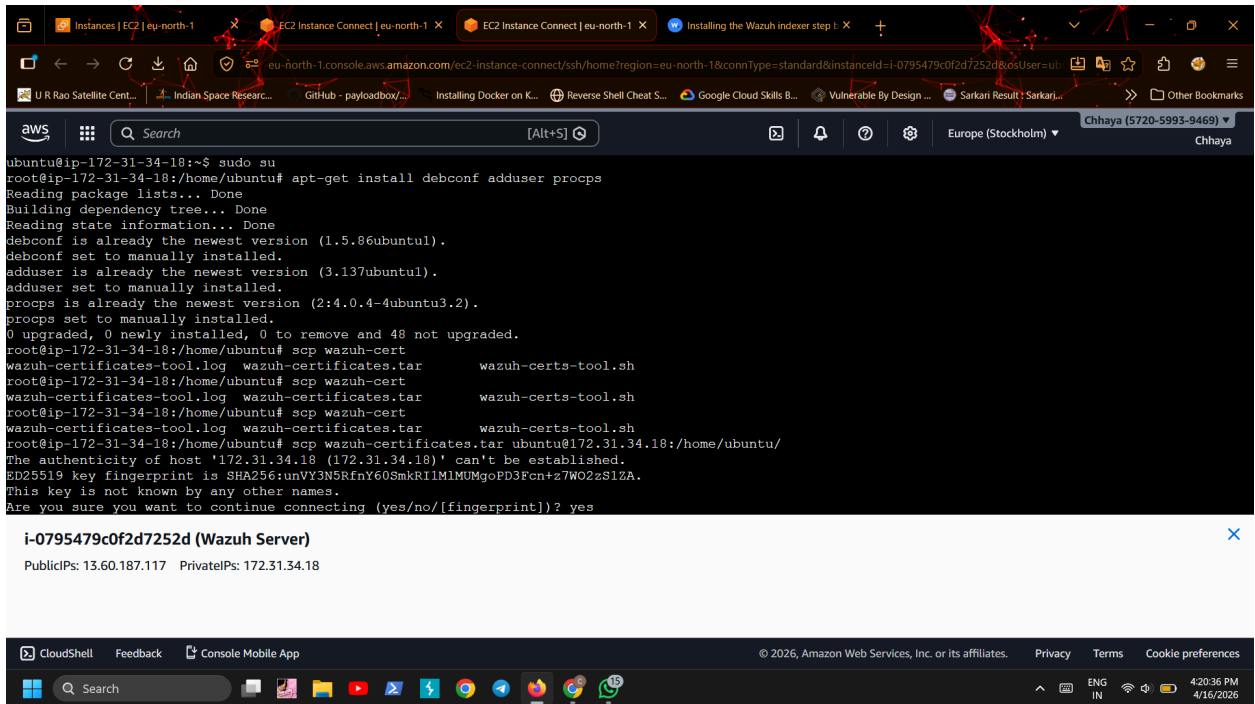
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3.2 Wazuh Indexer Installation

Next, I installed the Wazuh indexer on the Wazuh server.

- Installed required dependencies

- Added the Wazuh repository to the system
- Updated package list
- Installed the Wazuh indexer package



The screenshot shows a terminal window in AWS CloudShell. The user is logged in as root on an Ubuntu instance. The terminal output shows the following commands and their results:

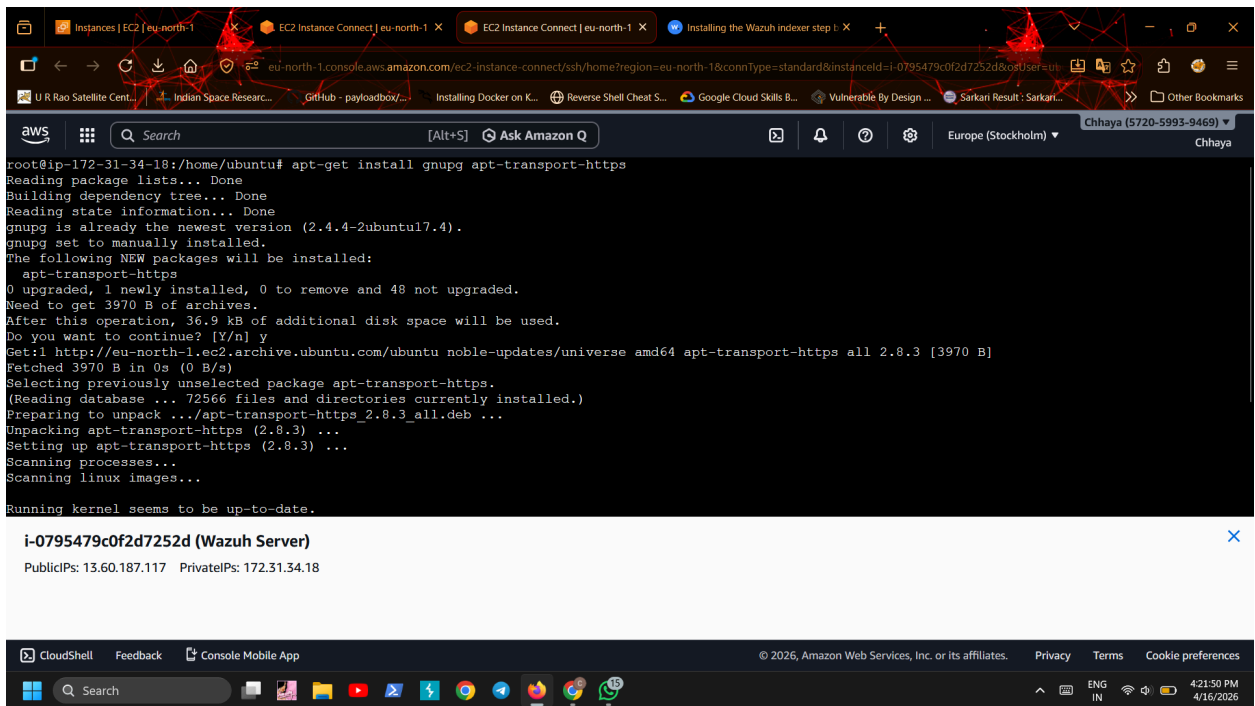
```

ubuntu@ip-172-31-34-18:~$ sudo su
root@ip-172-31-34-18:/home/ubuntu# apt-get install debconf adduser procs
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
debconf is already the newest version (1.5.86ubuntu1).
debconf set to manually installed.
adduser is already the newest version (3.137ubuntu1).
adduser set to manually installed.
procs is already the newest version (2:4.0.4-4ubuntu3.2).
procs set to manually installed.
0 upgraded, 0 newly installed, 0 to remove and 48 not upgraded.
root@ip-172-31-34-18:/home/ubuntu# scp wazuh-cert
wazuh-certificates-tool.log wazuh-certificates.tar wazuh-certs-tool.sh
root@ip-172-31-34-18:/home/ubuntu# scp wazuh-cert
wazuh-certificates-tool.log wazuh-certificates.tar wazuh-certs-tool.sh
root@ip-172-31-34-18:/home/ubuntu# scp wazuh-cert
wazuh-certificates-tool.log wazuh-certificates.tar wazuh-certs-tool.sh
root@ip-172-31-34-18:/home/ubuntu# scp wazuh-certificates.tar ubuntu@172.31.34.18:/home/ubuntu/
The authenticity of host '172.31.34.18 (172.31.34.18)' can't be established.
ED25519 key fingerprint is SHA256:unVY3N5RfnY60SmkRI1M1MUMgoPD3Fcn+z7WO2zS1ZA.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes

```

Below the terminal output, a notification box displays the instance ID and IP addresses:

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The screenshot shows a terminal window in AWS CloudShell. The user is logged in as root on an Ubuntu instance. The terminal output shows the following commands and their results:

```

root@ip-172-31-34-18:/home/ubuntu# apt-get install gnupg apt-transport-https
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
gnupg is already the newest version (2.4.4-2ubuntu17.4).
gnupg set to manually installed.
The following NEW packages will be installed:
  apt-transport-https
0 upgraded, 1 newly installed, 0 to remove and 48 not upgraded.
Need to get 3970 B of archives.
After this operation, 36.9 kB of additional disk space will be used.
Do you want to continue? [Y/n] y
Get:1 http://eu-north-1.ec2.archive.ubuntu.com/ubuntu noble-updates/universe amd64 apt-transport-https all 2.8.3 [3970 B]
Fetched 3970 B in 0s (0 B/s)
Selecting previously unselected package apt-transport-https.
(Reading database ... 72566 files and directories currently installed.)
Preparing to unpack .../apt-transport-https_2.8.3_all.deb ...
Unpacking apt-transport-https (2.8.3) ...
Setting up apt-transport-https (2.8.3) ...
Scanning processes...
Scanning linux images...
Running kernel seems to be up-to-date.

```

Below the terminal output, a notification box displays the instance ID and IP addresses:

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Scanning linux images...

Running kernel seems to be up-to-date.

No services need to be restarted.

No containers need to be restarted.

No user sessions are running outdated binaries.

No VM guests are running outdated hypervisor (qemu) binaries on this host.

```
root@ip-172-31-34-18:/home/ubuntu# curl -s https://packages.wazuh.com/key/GPG-KEY-WAZUH | gpg --no-default-keyring --keyring gnupg-ring:/usr/share/keyrings/wazuh.gpg --import && chmod 644 /usr/share/keyrings/wazuh.gpg
gpg: keyring '/usr/share/keyrings/wazuh.gpg' created
gpg: directory '/root/.gnupg' created
gpg: /root/.gnupg/trustdb.gpg: trustdb created
gpg: key 96B3EE5F29111145: public key "Wazuh.com (Wazuh Signing Key) <support@wazuh.com>" imported
gpg: Total number processed: 1
gpg:      imported: 1
root@ip-172-31-34-18:/home/ubuntu# echo "deb [signed-by=/usr/share/keyrings/wazuh.gpg] https://packages.wazuh.com/4.x/apt/ stable main" | tee -a /etc/apt/sources.list.d/wazuh.list
deb [signed-by=/usr/share/keyrings/wazuh.gpg] https://packages.wazuh.com/4.x/apt/ stable main
root@ip-172-31-34-18:/home/ubuntu#
```

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```
Fetches 68.6 kB in 19s (3581 B/s)
Reading package lists... Done
root@ip-172-31-34-18:/home/ubuntu# apt-get -y install wazuh-indexer
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following NEW packages will be installed:
  wazuh-indexer
0 upgraded, 1 newly installed, 0 to remove and 48 not upgraded.
Need to get 873 MB of archives.
After this operation, 1102 MB of additional disk space will be used.
Get:1 https://packages.wazuh.com/4.x/apt stable/main amd64 wazuh-indexer amd64 4.14.4-1 [873 MB]
Fetches 873 MB in 10s (86.2 MB/s)
Selecting previously unselected package wazuh-indexer.
(Reading database ... 72570 files and directories currently installed.)
Preparing to unpack .../wazuh-indexer_4.14.4-1_amd64.deb ...
Running Wazuh Indexer Pre-Installation Script
Unpacking wazuh-indexer (4.14.4-1) ...
Setting up wazuh-indexer (4.14.4-1) ...
Running Wazuh Indexer Post-Installation Script
### NOT starting on installation, please execute the following statements to configure wazuh-indexer service to start automatically using systemd
sudo systemctl daemon-reload
sudo systemctl enable wazuh-indexer.service
```

i-0795479c0f2d7252d (Wazuh Server)

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3.3 Indexer Configuration

After installation, I configured the indexer:

- Updated network settings (IP address of server)
- Set node name as defined earlier
- Configured cluster settings (single-node in my case)
- Adjusted discovery and node-related configurations

```
GNU nano 7.2 /etc/wazuh-indexer/openssl.yml *
network.host: "172.31.34.18"
node.name: "node-1"
cluster.initial_master_nodes:
- "node-1"
#- "node-2"
#- "node-3"
cluster.name: "wazuh-cluster"
#discovery.seed_hosts:
# - "node-1-ip"
# - "node-2-ip"
# - "node-3-ip"
node.max_local_storage_nodes: "3"
path.data: /var/lib/wazuh-indexer
path.logs: /var/log/wazuh-indexer

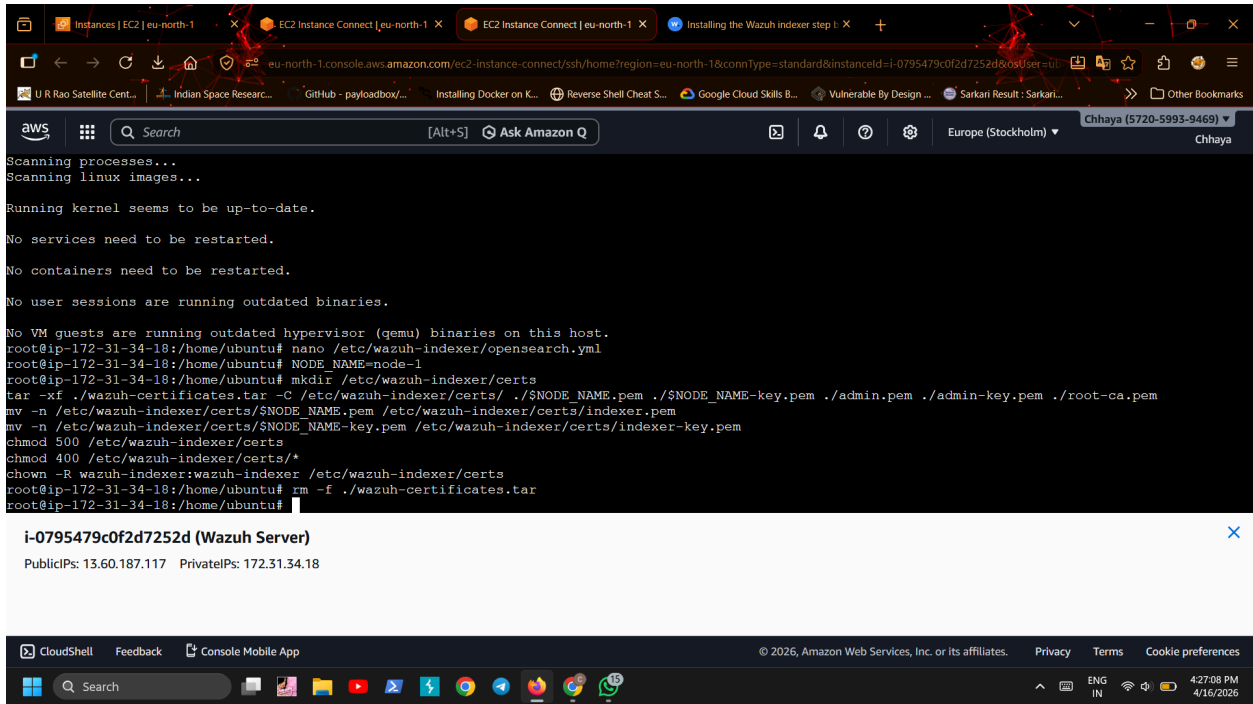
plugins.security.ssl.http.pemcert_filepath: /etc/wazuh-indexer/certs/indexer.pem
plugins.security.ssl.http.pemkey_filepath: /etc/wazuh-indexer/certs/indexer-key.pem
plugins.security.ssl.http.pemtrustedcas_filepath: /etc/wazuh-indexer/certs/root-ca.pem
plugins.security.ssl.transport.pemcert_filepath: /etc/wazuh-indexer/certs/indexer.pem

^G Help      ^O Write Out  ^W Where Is   ^X Cut        ^T Execute    ^C Location   ^M-U Undo      ^M-A Set Mark   ^M-] To Bracket
^X Exit      ^R Read File  ^_ Replace    ^U Paste      ^J Justify    ^/_ Go To Line  ^M-E Redo      ^M-C Copy    ^_ Where Was
```

i-0795479c0f2d7252d (Wazuh Server)
PublicIPs: 13.60.187.117 PrivateIPs: 172.31.34.18

3.4 Deploying Certificates

- Extracted the certificate file
- Moved required certificate files to the correct directory
- Renamed files as required by Wazuh
- Set proper permissions and ownership



The screenshot shows the AWS CloudShell interface with a terminal window. The terminal output indicates that the system is up-to-date and no services need to be restarted. The user is running a series of commands to install Wazuh, including creating certificates, setting permissions, and removing the installation tarball. The terminal output is as follows:

```
Scanning processes...
Scanning linux images...

Running kernel seems to be up-to-date.

No services need to be restarted.

No containers need to be restarted.

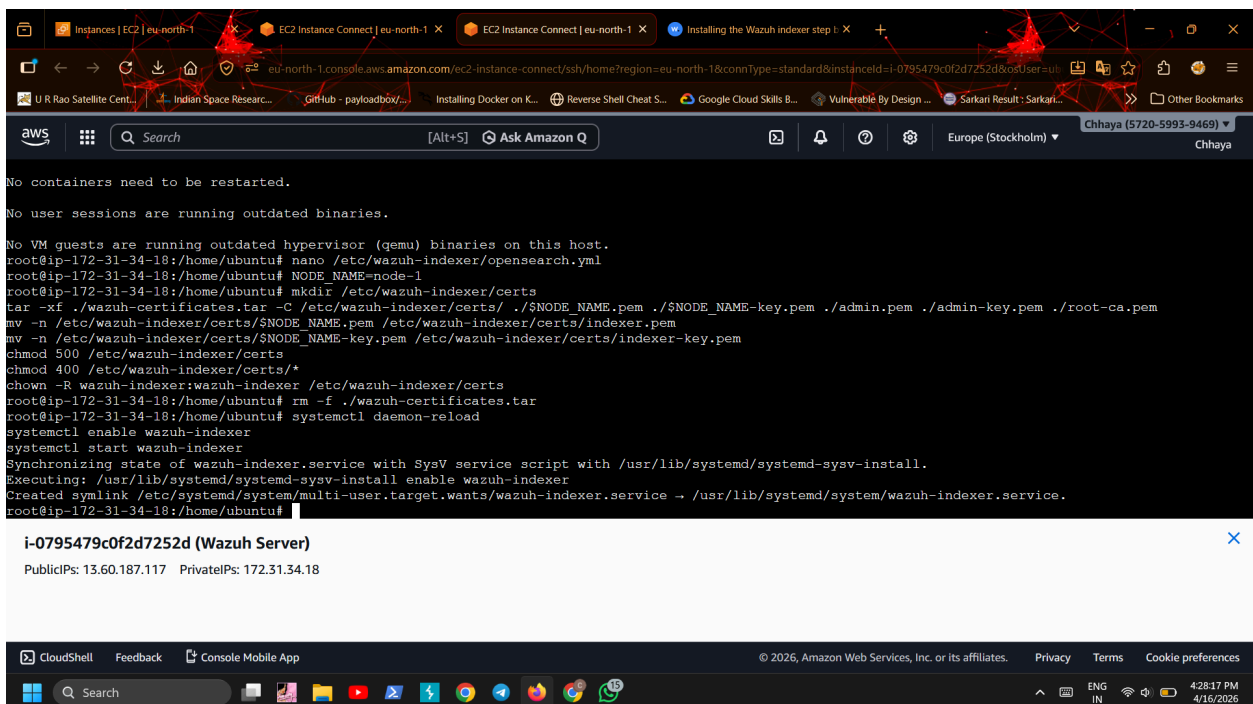
No user sessions are running outdated binaries.

No VM guests are running outdated hypervisor (qemu) binaries on this host.
root@ip-172-31-34-18:/home/ubuntu# nano /etc/wazuh-indexer/opensearch.yml
root@ip-172-31-34-18:/home/ubuntu# NODE_NAME=node-1
root@ip-172-31-34-18:/home/ubuntu# mkdir /etc/wazuh-indexer/certs
tar -xvf ./wazuh-certificates.tar -C /etc/wazuh-indexer/certs/ ./${NODE_NAME}.pem ./${NODE_NAME}-key.pem ./admin.pem ./admin-key.pem ./root-ca.pem
mv -n /etc/wazuh-indexer/certs/${NODE_NAME}.pem /etc/wazuh-indexer/certs/indexer.pem
mv -n /etc/wazuh-indexer/certs/${NODE_NAME}-key.pem /etc/wazuh-indexer/certs/indexer-key.pem
chmod 500 /etc/wazuh-indexer/certs
chmod 400 /etc/wazuh-indexer/certs/*
chown -R wazuh-indexer:wazuh-indexer /etc/wazuh-indexer/certs
root@ip-172-31-34-18:/home/ubuntu# rm -f ./wazuh-certificates.tar
root@ip-172-31-34-18:/home/ubuntu#
```

Below the terminal output, the instance details for **i-0795479c0f2d7252d (Wazuh Server)** are shown, including PublicIPs: 13.60.187.117 and PrivateIPs: 172.31.34.18.

3.5 Starting the Indexer Service

- Reloaded system services
- Enabled Wazuh indexer to start automatically
- Started the indexer service



The screenshot shows the AWS CloudShell interface with a terminal window. The terminal output indicates that the Wazuh indexer service is being started. The user is running a series of commands to enable the service and start it. The terminal output is as follows:

```
No containers need to be restarted.

No user sessions are running outdated binaries.

No VM guests are running outdated hypervisor (qemu) binaries on this host.
root@ip-172-31-34-18:/home/ubuntu# nano /etc/wazuh-indexer/opensearch.yml
root@ip-172-31-34-18:/home/ubuntu# NODE_NAME=node-1
root@ip-172-31-34-18:/home/ubuntu# mkdir /etc/wazuh-indexer/certs
tar -xvf ./wazuh-certificates.tar -C /etc/wazuh-indexer/certs/ ./${NODE_NAME}.pem ./${NODE_NAME}-key.pem ./admin.pem ./admin-key.pem ./root-ca.pem
mv -n /etc/wazuh-indexer/certs/${NODE_NAME}.pem /etc/wazuh-indexer/certs/indexer.pem
mv -n /etc/wazuh-indexer/certs/${NODE_NAME}-key.pem /etc/wazuh-indexer/certs/indexer-key.pem
chmod 500 /etc/wazuh-indexer/certs
chmod 400 /etc/wazuh-indexer/certs/*
chown -R wazuh-indexer:wazuh-indexer /etc/wazuh-indexer/certs
root@ip-172-31-34-18:/home/ubuntu# rm -f ./wazuh-certificates.tar
root@ip-172-31-34-18:/home/ubuntu# systemctl daemon-reload
systemctl enable wazuh-indexer
systemctl start wazuh-indexer
Synchronizing state of wazuh-indexer.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable wazuh-indexer
Created symlink /etc/systemd/system/multi-user.target.wants/wazuh-indexer.service -> /usr/lib/systemd/system/wazuh-indexer.service.
root@ip-172-31-34-18:/home/ubuntu#
```

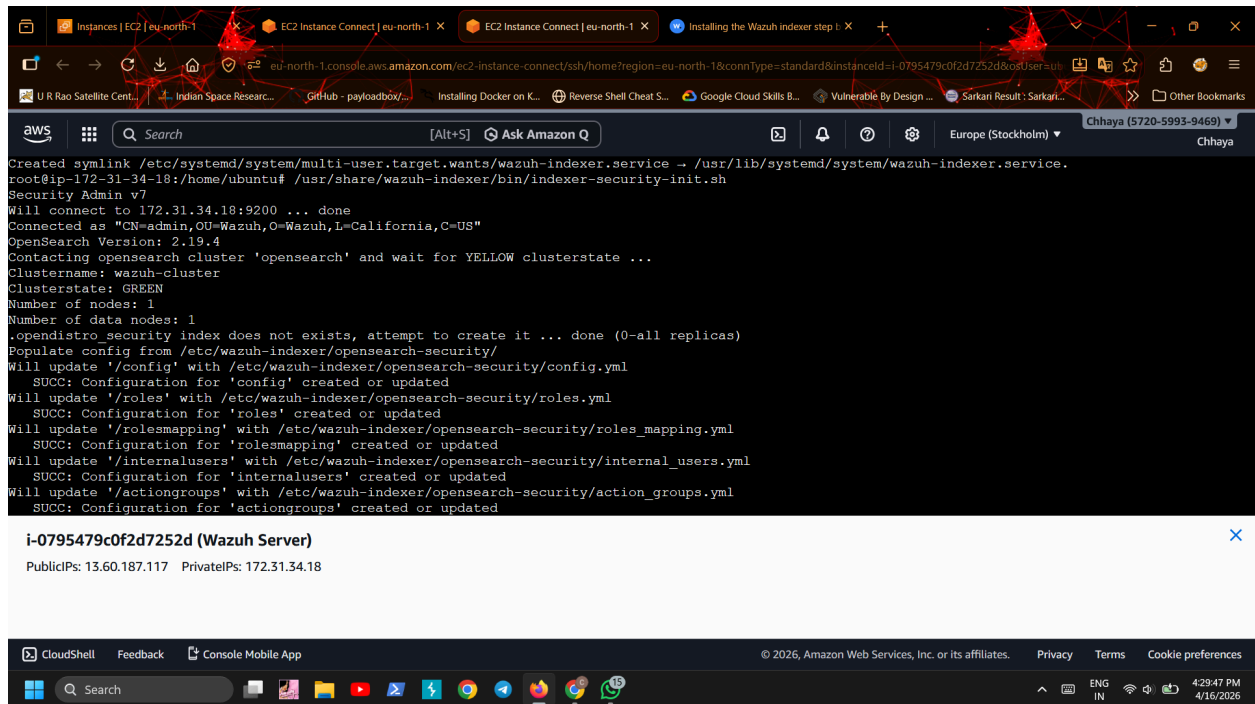
Below the terminal output, the instance details for **i-0795479c0f2d7252d (Wazuh Server)** are shown, including PublicIPs: 13.60.187.117 and PrivateIPs: 172.31.34.18.

3.6 Cluster Initialization

In the final step:

- Ran the initialization script
- This step loaded certificate information and started the cluster

Since I used a single-node setup, this was done only once.



```
Created symlink /etc/systemd/system/multi-user.target.wants/wazuh-indexer.service → /usr/lib/systemd/system/wazuh-indexer.service.
root@ip-172-31-34-18:/home/ubuntu# /usr/share/wazuh-indexer/bin/indexer-security-init.sh
Security Admin v7
Will connect to 172.31.34.18:9200 ... done
Connected as "CN=admin,OU=Wazuh,O=Wazuh,L=California,C=US"
OpenSearch Version: 2.19.4
Contacting opensearch cluster 'opensearch' and wait for YELLOW clusterstate ...
Clustername: wazuh-cluster
Clusterstate: GREEN
Number of nodes: 1
Number of data nodes: 1
.opendistro security index does not exists, attempt to create it ... done (0-all replicas)
Populate config from /etc/wazuh-indexer/opensearch-security/
Will update '/config' with /etc/wazuh-indexer/opensearch-security/config.yml
  SUCC: Configuration for 'config' created or updated
Will update '/roles' with /etc/wazuh-indexer/opensearch-security/roles.yml
  SUCC: Configuration for 'roles' created or updated
Will update '/rolesmapping' with /etc/wazuh-indexer/opensearch-security/roles_mapping.yml
  SUCC: Configuration for 'rolesmapping' created or updated
Will update '/internalusers' with /etc/wazuh-indexer/opensearch-security/internal_users.yml
  SUCC: Configuration for 'internalusers' created or updated
Will update '/actiongroups' with /etc/wazuh-indexer/opensearch-security/action_groups.yml
  SUCC: Configuration for 'actiongroups' created or updated
```

i-0795479c0f2d7252d (Wazuh Server)

PublicIPs: 13.60.187.117 PrivateIPs: 172.31.34.18

3.7 Testing the Installation

To verify the setup:

- Checked indexer response using API
- Confirmed that the node and cluster were working properly

```
Done with success
root@ip-172-31-34-18:/home/ubuntu# curl -k -u admin https://172.31.34.18:9200
Enter host password for user 'admin':
curl: (6) Could not resolve host: https
root@ip-172-31-34-18:/home/ubuntu# curl -k -u admin https://172.31.34.18:9200
Enter host password for user 'admin':
{
  "name" : "node-1",
  "cluster_name" : "wazuh-cluster",
  "cluster_uuid" : "xPYS8BLgRaKfWGg5Rm0B5A",
  "version" : {
    "number" : "7.10.2",
    "build_type" : "deb",
    "build_hash" : "eb914af4e4473af07d6b43e6140f86f07fa0d002",
    "build_date" : "2026-03-13T10:29:35.190091416Z",
    "build_snapshot" : false,
    "lucene_version" : "9.12.3",
    "minimum_wire_compatibility_version" : "7.10.0",
    "minimum_index_compatibility_version" : "7.0.0"
  },
  "tagline" : "The OpenSearch Project: https://opensearch.org/"
}
root@ip-172-31-34-18:/home/ubuntu#
```

i-0795479c0f2d7252d (Wazuh Server)

PublicIPs: 13.60.187.117 PrivateIPs: 172.31.34.18

```
root@ip-172-31-34-18:/home/ubuntu# curl -k -u admin https://172.31.34.18:9200
Enter host password for user 'admin':
{
  "name" : "node-1",
  "cluster_name" : "wazuh-cluster",
  "cluster_uuid" : "xPYS8BLgRaKfWGg5Rm0B5A",
  "version" : {
    "number" : "7.10.2",
    "build_type" : "deb",
    "build_hash" : "eb914af4e4473af07d6b43e6140f86f07fa0d002",
    "build_date" : "2026-03-13T10:29:35.190091416Z",
    "build_snapshot" : false,
    "lucene_version" : "9.12.3",
    "minimum_wire_compatibility_version" : "7.10.0",
    "minimum_index_compatibility_version" : "7.0.0"
  },
  "tagline" : "The OpenSearch Project: https://opensearch.org/"
}
root@ip-172-31-34-18:/home/ubuntu# curl -k -u admin https://172.31.34.18:9200/_cat/nodes?v
Enter host password for user 'admin':
ip heap.percent ram.percent cpu load 1m load 5m load 15m node.role node.roles cluster_manager name
172.31.34.18 22 54 3 0.03 0.08 0.03 dimr cluster_manager,data,ingest,remote_cluster_client * node-1
root@ip-172-31-34-18:/home/ubuntu#
```

i-0795479c0f2d7252d (Wazuh Server)

PublicIPs: 13.60.187.117 PrivateIPs: 172.31.34.18

4. Wazuh Server Installation (Wazuh Server Machine)

After setting up the Wazuh indexer, the next step was to install the Wazuh server on the Wazuh server machine. This component is responsible for collecting logs from agents and generating alerts.

The installation was done in stages as per the official documentation.

4.1 Adding Wazuh Repository

First, I configured the Wazuh repository on the system.

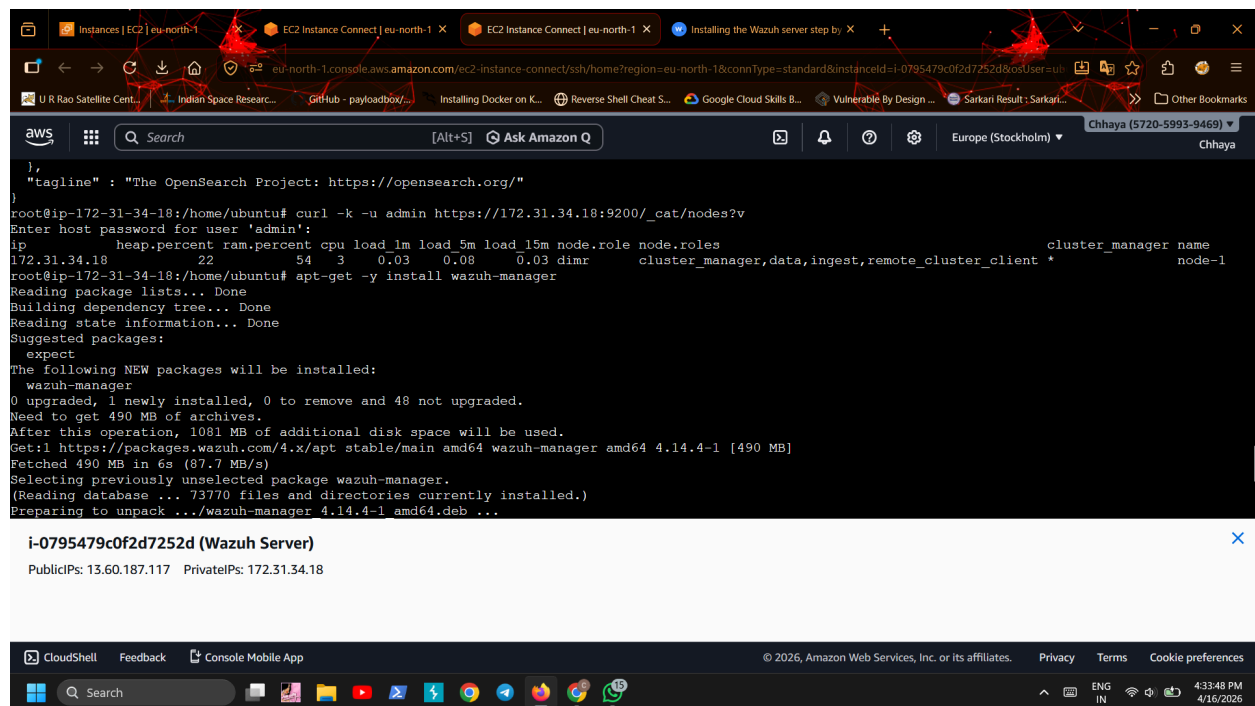
- Installed required packages for secure repository handling
- Added the Wazuh GPG key
- Added the official Wazuh repository
- Updated the system package list

4.2 Installing Wazuh Manager

Next, I installed the Wazuh manager.

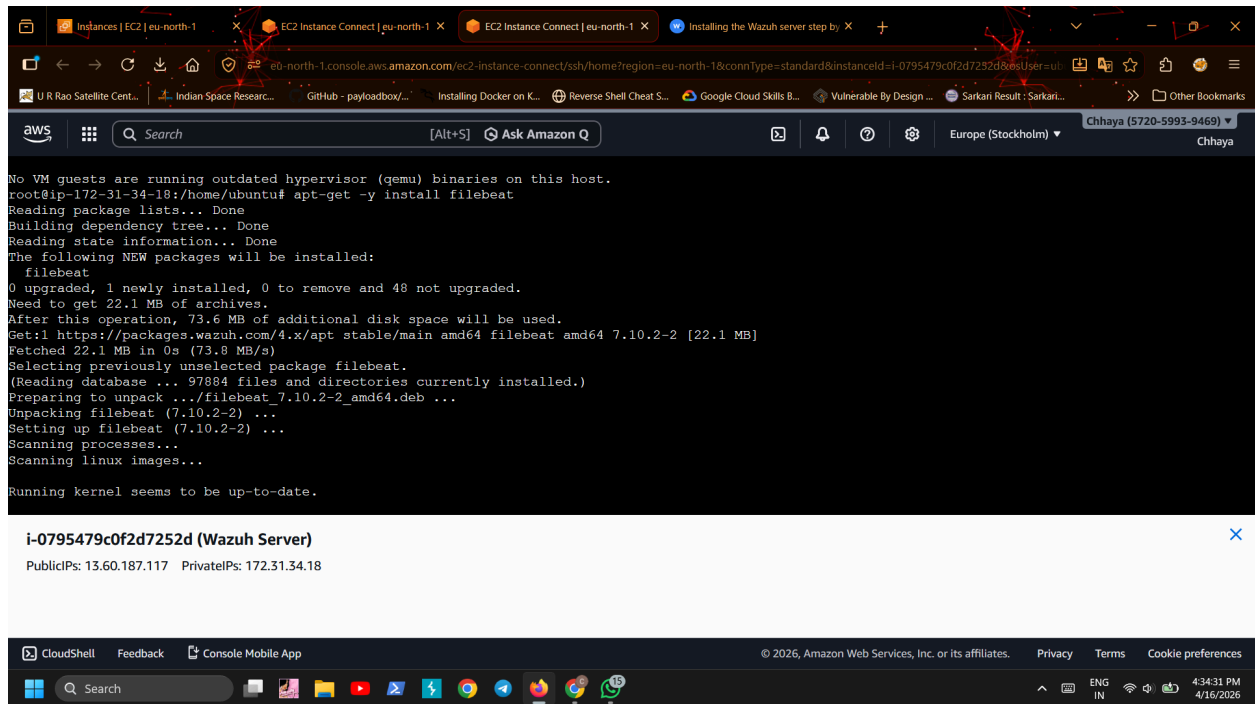
- Installed the manager package
- This component is responsible for collecting and analyzing logs from the Wazuh Agent

After installation, the manager was ready for configuration.



```
},
  "tagline" : "The OpenSearch Project: https://opensearch.org/"
}
root@ip-172-31-34-18:/home/ubuntu# curl -k -u admin https://172.31.34.18:9200/_cat/nodes?v
Enter host password for user 'admin':
ip      heap.percent ram.percent cpu load_1m load_5m load_15m node.role node.roles
172.31.34.18      22      54      3      0.03      0.08      0.03 dimr      cluster_manager,data,ingest,remote_cluster_client *
root@ip-172-31-34-18:/home/ubuntu# apt-get -y install wazuh-manager
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
Suggested packages:
  expect
The following NEW packages will be installed:
  wazuh-manager
0 upgraded, 1 newly installed, 0 to remove and 48 not upgraded.
Need to get 490 MB of archives.
After this operation, 1081 MB of additional disk space will be used.
Get:1 https://packages.wazuh.com/4.x/apt stable/main amd64 wazuh-manager amd64 4.14.4-1 [490 MB]
Fetched 490 MB in 6s (87.7 MB/s)
Selecting previously unselected package wazuh-manager.
(Reading database ... 73770 files and directories currently installed.)
Preparing to unpack .../wazuh-manager 4.14.4-1 amd64.deb ...

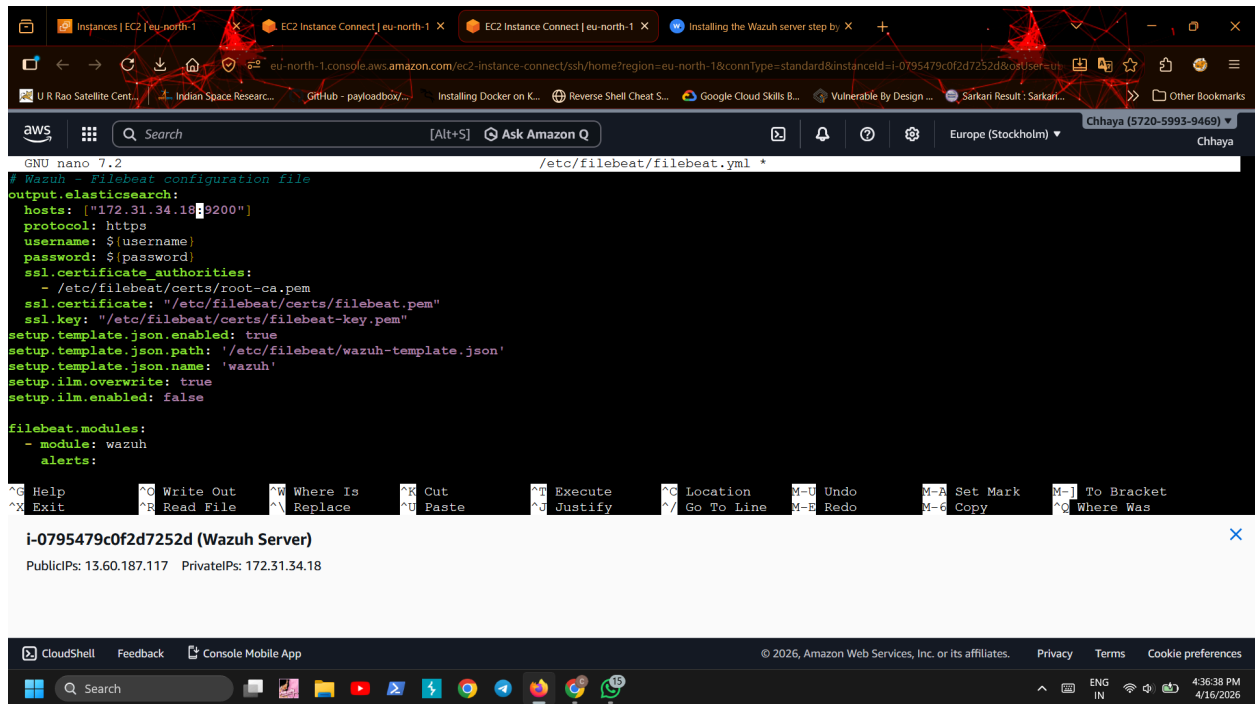
i-0795479c0f2d7252d (Wazuh Server)
PublicIPs: 13.60.187.117 PrivateIPs: 172.31.34.18
```

4.4 Configuring Filebeat

Then I configured Filebeat to connect with the Wazuh indexer.

- Downloaded the preconfigured Filebeat configuration file
- Updated the indexer IP address (replaced default localhost with actual IP of Wazuh indexer)
- Configured authentication credentials
- Set up secure communication using certificates



4.5 Setting Up Filebeat Keystore

To store credentials securely:

- Created Filebeat keystore
- Added username and password (default admin credentials)

This ensures sensitive data is not stored in plain text.

Instances | EC2 | eu-north-1

EC2 Instance Connect | eu-north-1

EC2 Instance Connect | eu-north-1

Installing the Wazuh server step by

eu-north-1.console.aws.amazon.com/ec2-instance-connect/ssh/home?region=eu-north-1&connType=standard&instanceId=i-0795479c0f2d7252d&osUser=ubuntu

U R Rao Satellite Cent... Indian Space Researc... GitHub - payloadbox/... Installing Docker on K... Reverse Shell Cheat S... Google Cloud Skills B... Vulnerable By Design ... Sarkari Result: Sarkari...

aws

Search

[Alt+S] Ask Amazon Q

Europe (Stockholm)

Chhaya (5720-5993-9469)

Chhaya

No user sessions are running outdated binaries.

No VM guests are running outdated hypervisor (qemu) binaries on this host.

root@ip-172-31-34-18:/home/ubuntu# curl -sO /etc/filebeat/filebeat.yml https://packages.wazuh.com/4.14/tpl/wazuh/filebeat/filebeat.yml

root@ip-172-31-34-18:/home/ubuntu# /etc/filebeat/filebeat.yml https://packages.wazuh.com/4.14/tpl/wazuh/filebeat/filebeat.yml

bash: /etc/filebeat/filebeat.yml: Permission denied

root@ip-172-31-34-18:/home/ubuntu# nano /etc/filebeat/filebeat.yml

root@ip-172-31-34-18:/home/ubuntu# nano /etc/filebeat/filebeat.yml

root@ip-172-31-34-18:/home/ubuntu# filebeat keystore create

Created filebeat keystore

root@ip-172-31-34-18:/home/ubuntu# echo admin | filebeat keystore add username --stdin --force

echo admin | filebeat keystore add password --stdin --force

Successfully updated the keystore

Successfully updated the keystore

root@ip-172-31-34-18:/home/ubuntu# curl -sO /etc/filebeat/wazuh-template.json https://raw.githubusercontent.com/wazuh/wazuh/v4.14.4/extensions/elasticse

arch/7.x/wazuh-template.json

chmod go+r /etc/filebeat/wazuh-template.json

root@ip-172-31-34-18:/home/ubuntu# curl -s https://packages.wazuh.com/4.x/filebeat/wazuh-filebeat-0.5.tar.gz | tar -xvz -C /usr/share/filebeat/module

wazuh/

wazuh/alerts/

wazuh/alerts/manifest.yml

wazuh/alerts/config/

CloudShell Feedback Console Mobile App

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Search

4:37:48 PM 4/16/2026

i-0795479c0f2d7252d (Wazuh Server)

PublicIPs: 13.60.187.117 PrivateIPs: 172.31.34.18

Instances | EC2 | eu-north-1

EC2 Instance Connect | eu-north-1

EC2 Instance Connect | eu-north-1

Installing the Wazuh server step by

eu-north-1.console.aws.amazon.com/ec2-instance-connect/ssh/home?region=eu-north-1&connType=standard&instanceId=i-0795479c0f2d7252d&osUser=ubuntu

U R Rao Satellite Cent... Indian Space Researc... GitHub - payloadbox/... Installing Docker on K... Reverse Shell Cheat S... Google Cloud Skills B... Vulnerable By Design ... Sarkari Result: Sarkari...

aws

Search

[Alt+S] Ask Amazon Q

Europe (Stockholm)

Chhaya (5720-5993-9469)

Chhaya

root@ip-172-31-34-18:/home/ubuntu# curl -sO /etc/filebeat/wazuh-template.json https://raw.githubusercontent.com/wazuh/wazuh/v4.14.4/extensions/elasticse

arch/7.x/wazuh-template.json

chmod go+r /etc/filebeat/wazuh-template.json

root@ip-172-31-34-18:/home/ubuntu# curl -s https://packages.wazuh.com/4.x/filebeat/wazuh-filebeat-0.5.tar.gz | tar -xvz -C /usr/share/filebeat/module

wazuh/

wazuh/alerts/

wazuh/alerts/manifest.yml

wazuh/alerts/config/

wazuh/alerts/config/alerts.yml

wazuh/alerts/ingest/

wazuh/alerts/ingest/pipeline.json

wazuh/archives/

wazuh/archives/manifest.yml

wazuh/archives/config/

wazuh/archives/config/archives.yml

wazuh/archives/ingest/

wazuh/archives/ingest/pipeline.json

wazuh/_meta/

wazuh/_meta/config.yml

wazuh/_meta/fields.yml

wazuh/_meta/docs.asciidoc

wazuh/module.yml

root@ip-172-31-34-18:/home/ubuntu#

CloudShell Feedback Console Mobile App

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Search

4:38:30 PM 4/16/2026

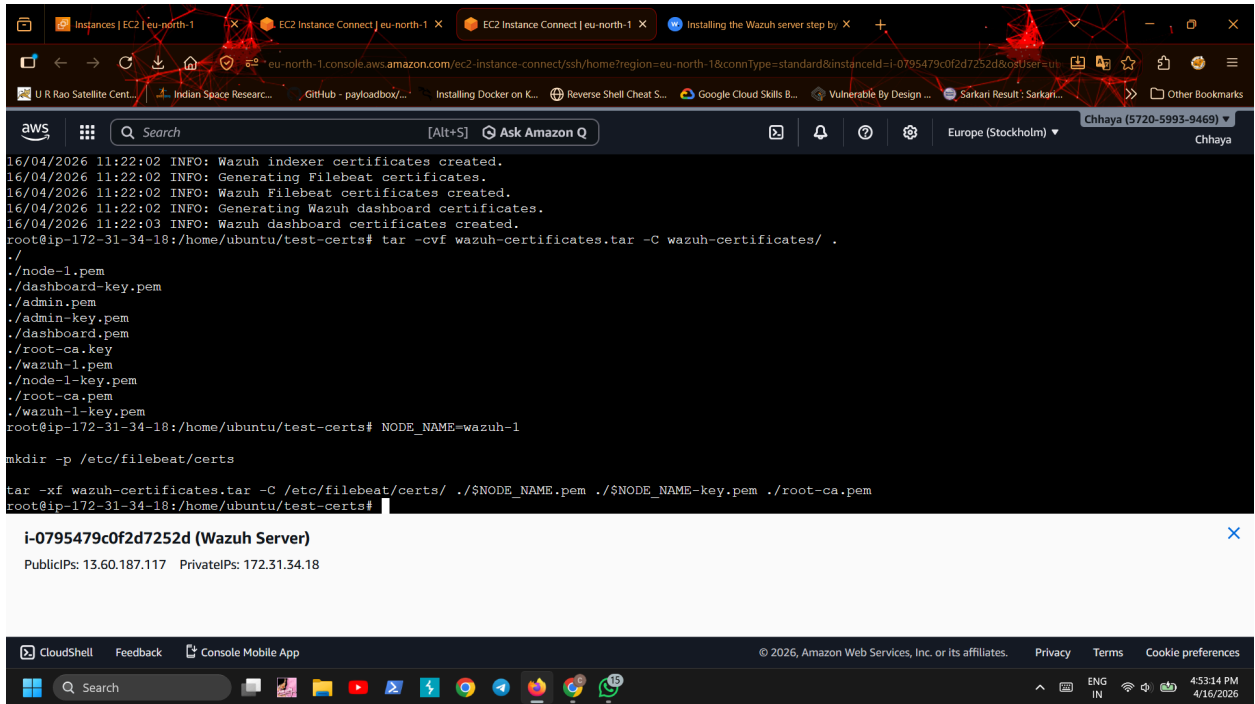
i-0795479c0f2d7252d (Wazuh Server)

PublicIPs: 13.60.187.117 PrivateIPs: 172.31.34.18

4.6 Deploying Certificates

After that, I deployed certificates for secure communication.

- Extracted certificates generated earlier
- Placed them in the correct Filebeat directory
- Renamed files as required
- Set proper permissions



```
16/04/2026 11:22:02 INFO: Wazuh indexer certificates created.
16/04/2026 11:22:02 INFO: Generating Filebeat certificates.
16/04/2026 11:22:02 INFO: Wazuh Filebeat certificates created.
16/04/2026 11:22:02 INFO: Generating Wazuh dashboard certificates.
16/04/2026 11:22:03 INFO: Wazuh dashboard certificates created.
root@ip-172-31-34-18:/home/ubuntu/test-certs# tar -cvf wazuh-certificates.tar -C wazuh-certificates/ .
./
./node-1.pem
./dashboard-key.pem
./admin.pem
./admin-key.pem
./dashboard.pem
./root-ca.key
./wazuh-1.pem
./node-1-key.pem
./root-ca.pem
./wazuh-1-key.pem
root@ip-172-31-34-18:/home/ubuntu/test-certs# NODE_NAME=wazuh-1

mkdir -p /etc/filebeat/certs

tar -xvf wazuh-certificates.tar -C /etc/filebeat/certs/ ./.$NODE_NAME.pem ./.$NODE_NAME-key.pem ./root-ca.pem
root@ip-172-31-34-18:/home/ubuntu/test-certs#
```

i-0795479c0f2d7252d (Wazuh Server)

PublicIPs: 13.60.187.117 PrivateIPs: 172.31.34.18

4.7 Configuring Indexer Connection in Wazuh Server

Next, I configured the connection between Wazuh server and indexer.

- Stored indexer credentials in Wazuh keystore
- Updated configuration file to include indexer IP address
- Verified certificate paths

4.8 Starting Wazuh Manager Service

After configuration:

- Enabled and started Wazuh manager service
- Verified that the service was running properly

The screenshot shows the AWS CloudShell interface. The terminal output displays the status of the `wazuh-manager.service` as `loaded` and `active (running)`. It also lists the processes running under the `wazuh-manager.service` CGroup, including `wazuh_apid.py`, `wazuh-authd`, `wazuh-db`, `wazuh-execd`, `wazuh-analysisd`, `wazuh-syscheckd`, `wazuh-remoted`, `wazuh-logcollector`, `wazuh-monitord`, and `wazuh-modulesd`. Below the terminal output, the instance ID `i-0795479c0f2d7252d` is shown, along with its public and private IP addresses. The bottom of the screenshot shows the AWS CloudShell footer with navigation links and system information.

```
wazuh-manager.service - Wazuh manager
Loaded: loaded (/usr/lib/systemd/system/wazuh-manager.service; enabled; preset: enabled)
Active: active (running) since Thu 2026-04-16 11:28:26 UTC; 45s ago
Process: 53460 ExecStart=/usr/bin/env /var/ossec/bin/wazuh-control start (code=exited, status=0/SUCCESS)
Tasks: 218 (limit: 9279)
Memory: 1.6G (peak: 1.6G)
CPU: 22.261s
CGroup: /system.slice/wazuh-manager.service
├─53523 /var/ossec/framework/python/bin/python3 /var/ossec/api/scripts/wazuh_apid.py
├─53562 /var/ossec/bin/wazuh-authd
├─53577 /var/ossec/bin/wazuh-db
├─53603 /var/ossec/framework/python/bin/python3 /var/ossec/api/scripts/wazuh_apid.py
├─53604 /var/ossec/framework/python/bin/python3 /var/ossec/api/scripts/wazuh_apid.py
├─53607 /var/ossec/framework/python/bin/python3 /var/ossec/api/scripts/wazuh_apid.py
├─53610 /var/ossec/framework/python/bin/python3 /var/ossec/api/scripts/wazuh_apid.py
├─53622 /var/ossec/bin/wazuh-execd
├─53634 /var/ossec/bin/wazuh-analysisd
├─53676 /var/ossec/bin/wazuh-syscheckd
├─53697 /var/ossec/bin/wazuh-remoted
├─53732 /var/ossec/bin/wazuh-logcollector
├─53748 /var/ossec/bin/wazuh-monitord
└─53757 /var/ossec/bin/wazuh-modulesd
```

lines 1-22

i-0795479c0f2d7252d (Wazuh Server)

PublicIPs: 13.60.187.117 PrivateIPs: 172.31.34.18

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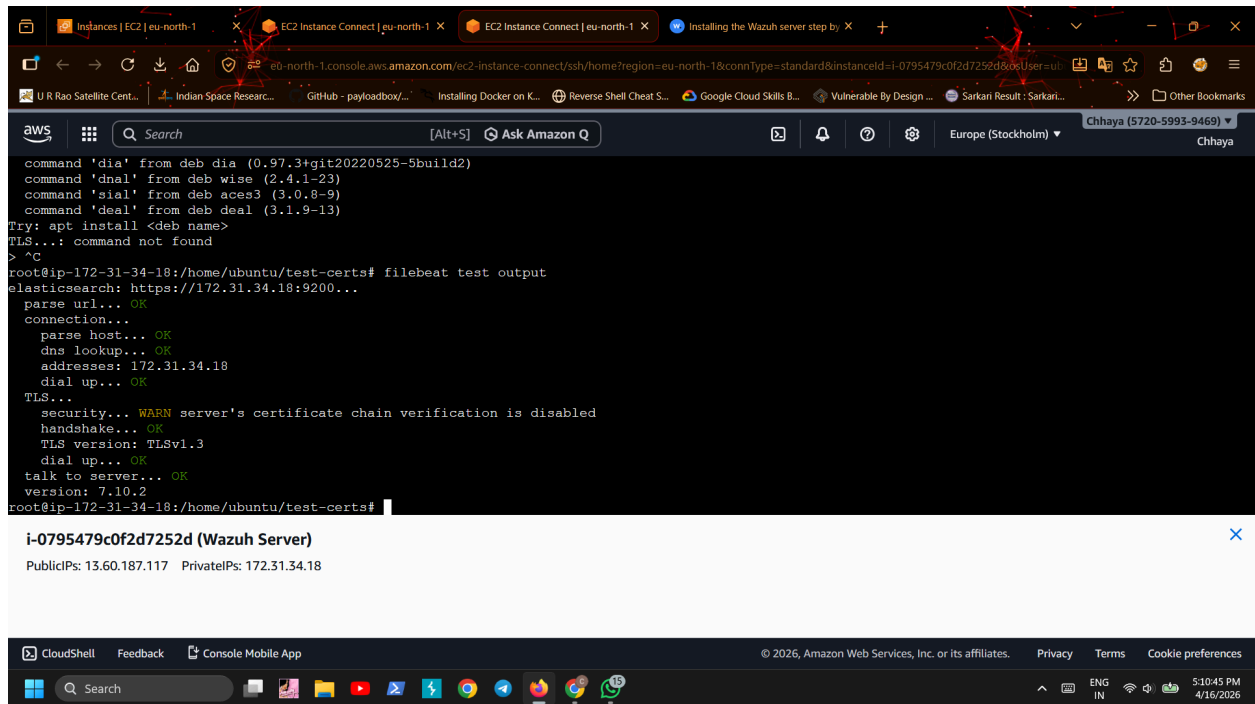
Search ENG IN 4:59:19 PM 4/16/2026

4.9 Starting Filebeat Service

Then I started Filebeat:

- Enabled and started Filebeat service
- Tested the connection to ensure logs are being forwarded correctly

The output confirmed successful communication with the indexer.

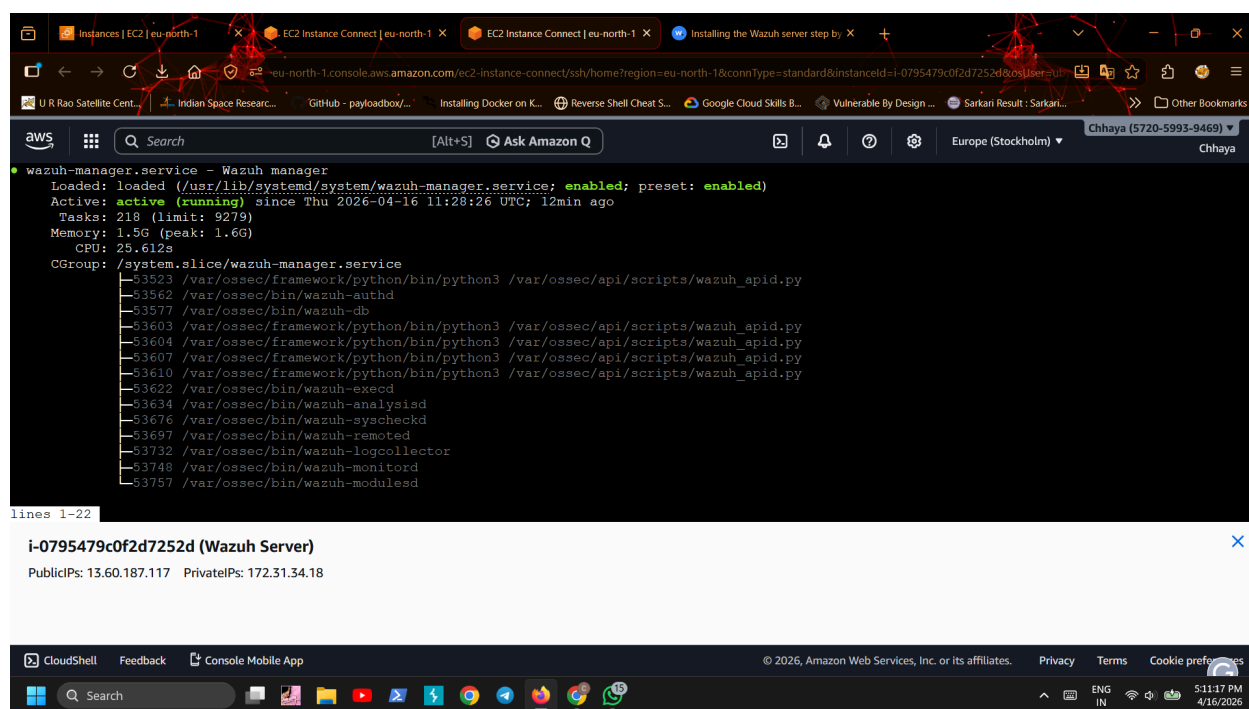


4.10 Verification

After completing all steps:

- Wazuh manager was running successfully
- Filebeat was forwarding logs to indexer
- No errors were observed in service status

This confirmed that the Wazuh server setup was completed successfully.



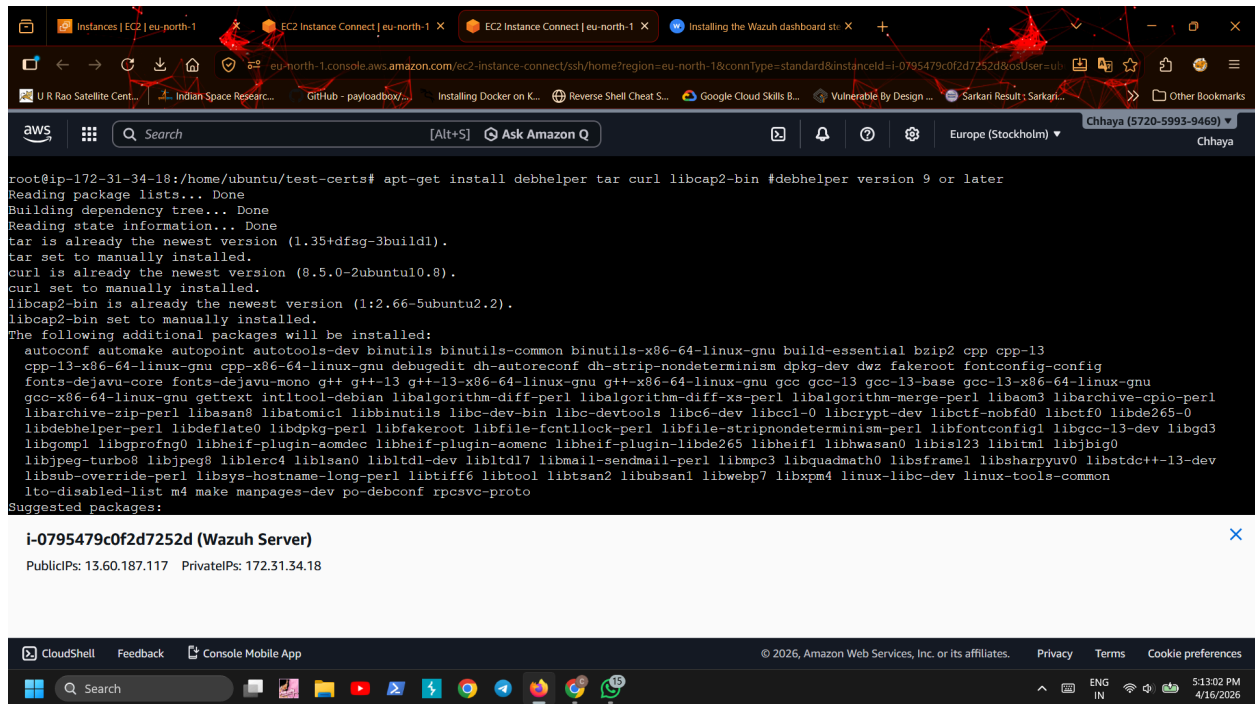
5. Wazuh Dashboard Installation (Wazuh Server Machine)

After completing the installation of the Wazuh indexer and Wazuh server, the final component installed was the Wazuh Dashboard. This is the web interface used to visualize alerts, monitor agents, and analyze security events.

5.1 Installing Required Dependencies

First, I installed the required system packages needed for the dashboard installation.

These packages help in handling archives, downloads, and system utilities required during setup.



5.2 Adding Wazuh Repository

Next, I configured the Wazuh repository on the system.

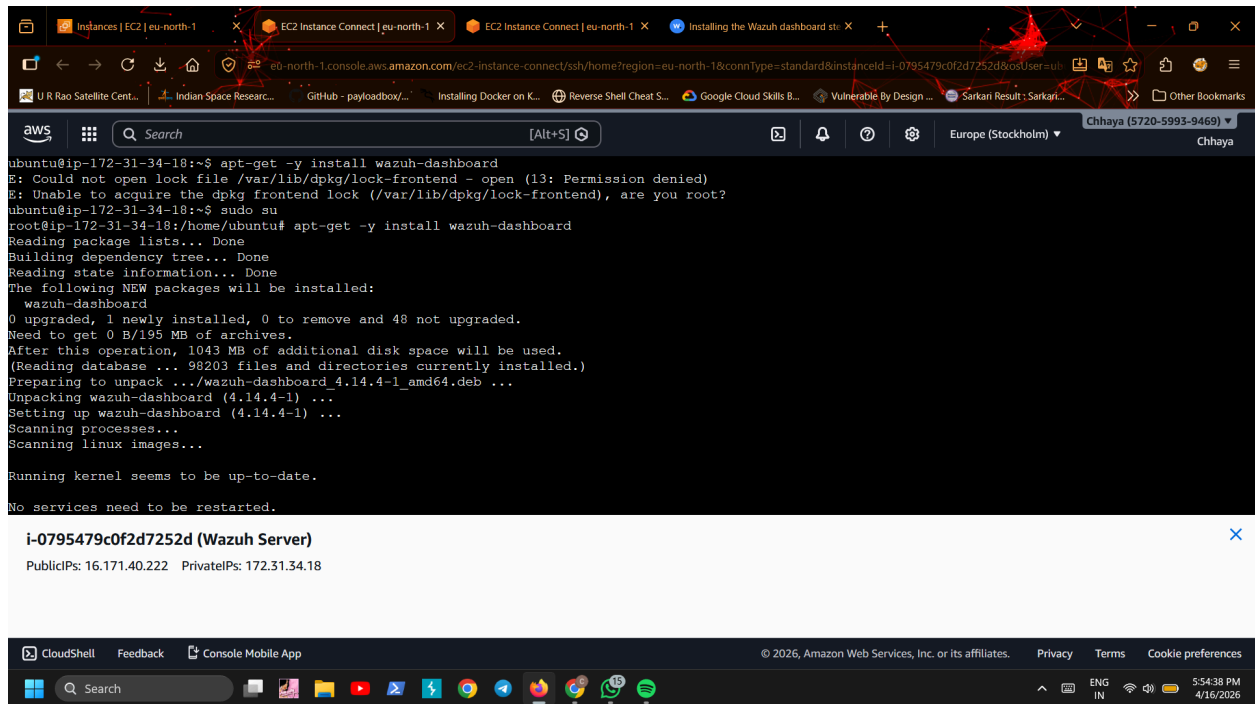
- Installed required repository tools
- Added the Wazuh GPG key for secure package verification
- Added the official Wazuh repository
- Updated system package list

This ensures that the latest Wazuh dashboard package can be installed.

5.3 Installing Wazuh Dashboard

After repository setup:

- Installed the Wazuh dashboard package
- This provides the web-based interface for monitoring logs and alerts

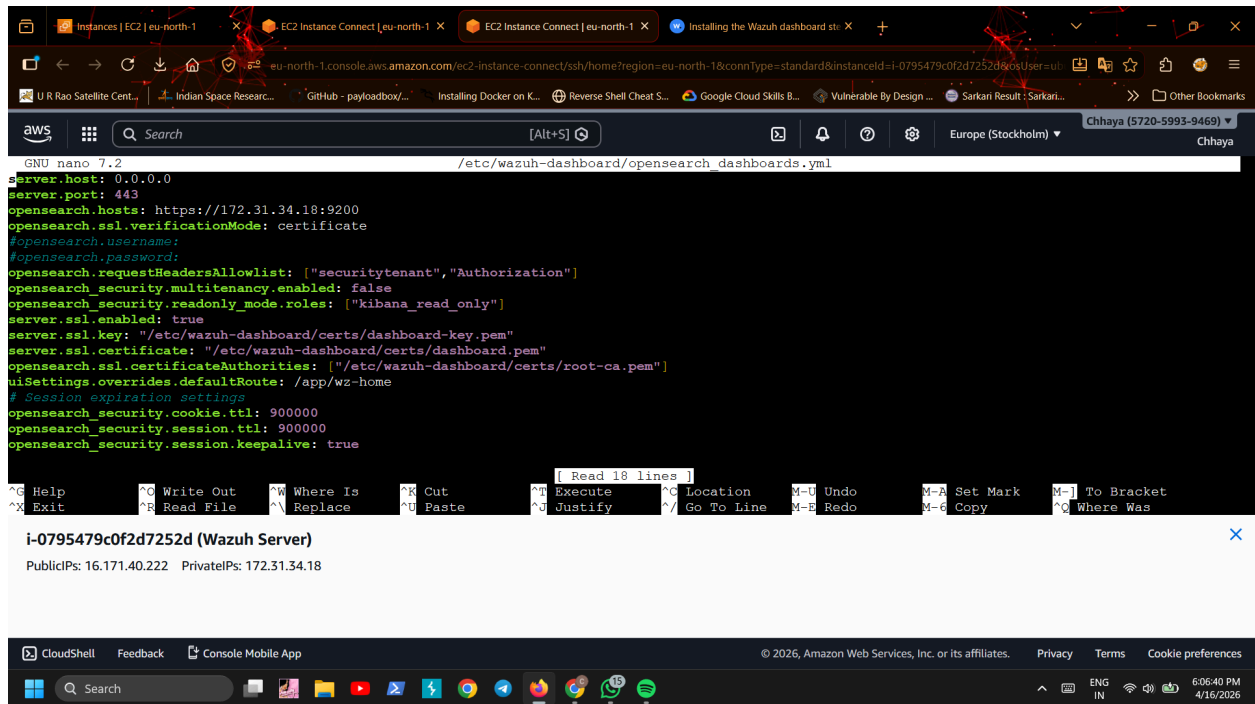


5.4 Configuring Wazuh Dashboard

After installation, I configured the dashboard settings.

- Set dashboard to listen on all available network interfaces
- Configured port settings for secure access
- Added Wazuh indexer connection details
- Disabled SSL verification mode for local setup (as per lab environment)

This step ensures the dashboard can communicate with the indexer properly.



5.5 Deploying Certificates

Next, I deployed SSL certificates for secure communication.

- Extracted certificates generated earlier during installation
- Copied them into the Wazuh dashboard certificate directory
- Renamed certificate files as required
- Set proper permissions and ownership

This ensures encrypted communication between the dashboard and other components.

The screenshot shows an AWS CloudShell terminal window with the following commands and output:

```
mkdir -p /etc/wazuh-dashboard/certs

tar -xvf wazuh-certificates.tar -C /etc/wazuh-dashboard/certs/ \
./$NODE_NAME.pem ./.$NODE_NAME-key.pem ./root-ca.pem
tar: This does not look like a tar archive
tar: ./dashboard.pem: Not found in archive
tar: ./dashboard-key.pem: Not found in archive
tar: ./root-ca.pem: Not found in archive
tar: Exiting with failure status due to previous errors
root@ip-172-31-34-18:/home/ubuntu# cd test-certs/
root@ip-172-31-34-18:/home/ubuntu/test-certs# NODE_NAME=dashboard

mkdir -p /etc/wazuh-dashboard/certs

tar -xvf wazuh-certificates.tar -C /etc/wazuh-dashboard/certs/ \
./$NODE_NAME.pem ./.$NODE_NAME-key.pem ./root-ca.pem
root@ip-172-31-34-18:/home/ubuntu/test-certs# mv /etc/wazuh-dashboard/certs/dashboard.pem /etc/wazuh-dashboard/certs/dashboard.pem 2>/dev/null
mv /etc/wazuh-dashboard/certs/dashboard-key.pem /etc/wazuh-dashboard/certs/dashboard-key.pem 2>/dev/null
root@ip-172-31-34-18:/home/ubuntu/test-certs# chmod 500 /etc/wazuh-dashboard/certs
chmod 400 /etc/wazuh-dashboard/certs/*
chown -R wazuh-dashboard:wazuh-dashboard /etc/wazuh-dashboard/certs
root@ip-172-31-34-18:/home/ubuntu/test-certs#
```

A notification for the Wazuh Server instance is displayed below the terminal output:

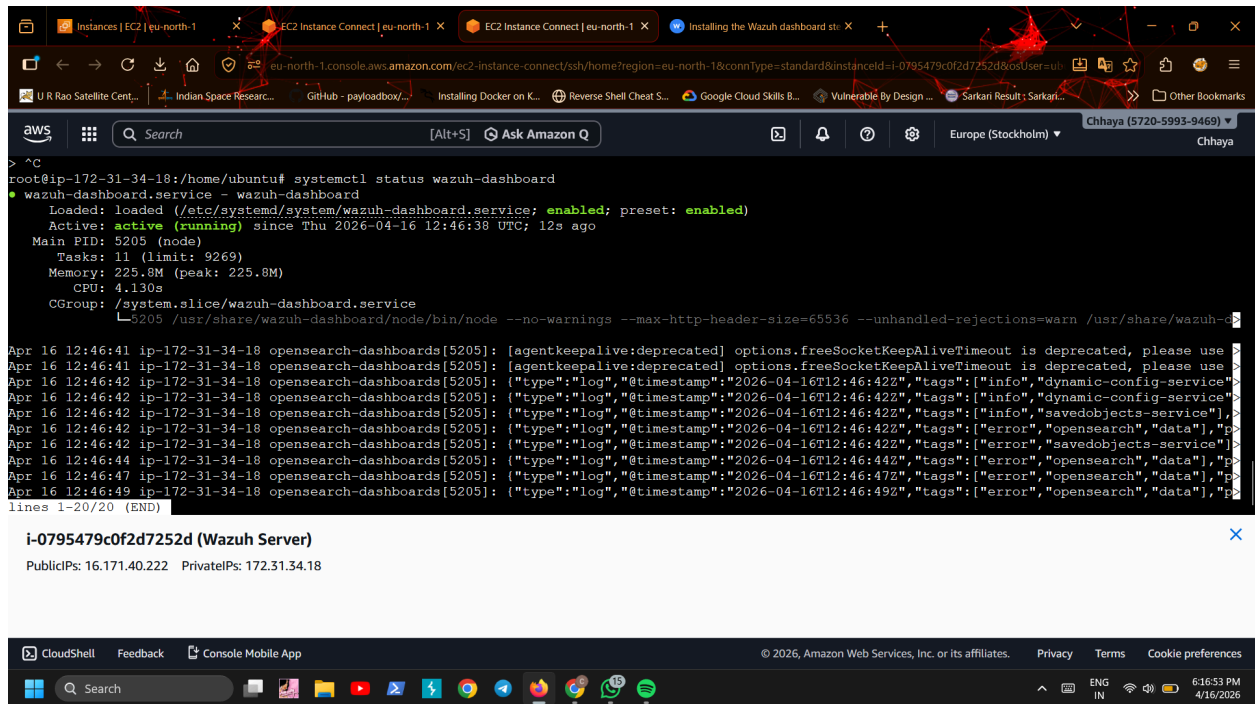
i-0795479c0f2d7252d (Wazuh Server)
PublicIPs: 16.171.40.222 PrivateIPs: 172.31.34.18

The bottom of the screenshot shows the AWS CloudShell interface with the CloudShell logo, Feedback, Console Mobile App, and a footer with the Amazon Web Services logo, copyright notice, and links to Privacy, Terms, and Cookie preferences.

5.6 Starting Wazuh Dashboard Service

After configuration:

- Enabled Wazuh dashboard service
- Started the service using system controls
- Verified that the dashboard was running successfully



5.7 Connecting Dashboard to Wazuh Server

Then I configured the connection between the dashboard the Wazuh server:

- Updated Wazuh server IP in dashboard configuration file
- Used default API credentials for connection
- Enabled secure communication between dashboard and server

This allows the dashboard to fetch alerts and logs from the Wazuh server.

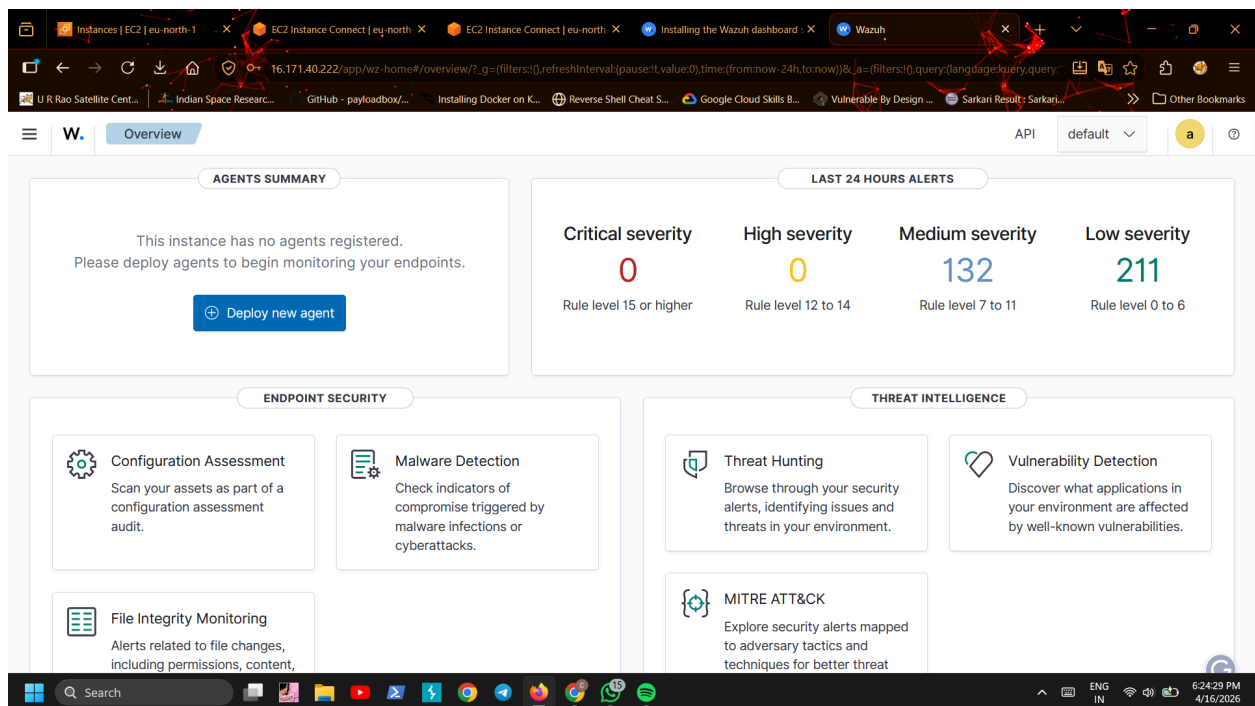
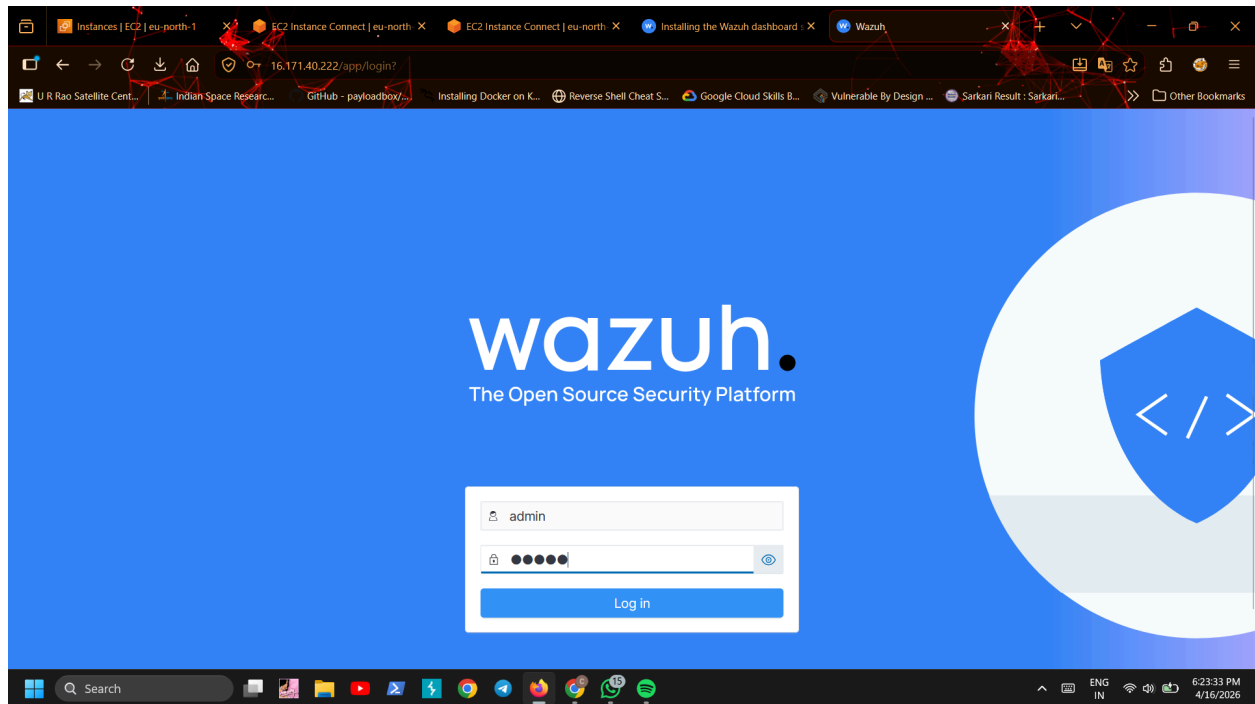
5.8 Accessing Wazuh Dashboard

After setup, I accessed the Wazuh web interface:

- Opened dashboard using browser
- Used default credentials:
 - Username: admin
 - Password: admin

On first login, the browser showed a certificate warning due to self-signed certificates. This was expected in a local lab setup and was accepted manually.

After login, the Wazuh dashboard loaded successfully.



5.9 Disabling Updates (Optional Step)

After completing installation, I disabled automatic repository updates to avoid accidental upgrades that could break the setup.

5.10 Securing Installation (Important Step)

Finally, I updated default credentials for security purposes.

- Changed internal user passwords using Wazuh security tool
- Updated API and indexer credentials
- Restarted services to apply changes

This step improves security of the entire Wazuh environment.

6. Wazuh Agent Installation (Wazuh Agent Machine)

After completing the setup of the Wazuh Server, Indexer, and Dashboard, the next step was to install the Wazuh Agent on a separate virtual machine named Wazuh Agent.

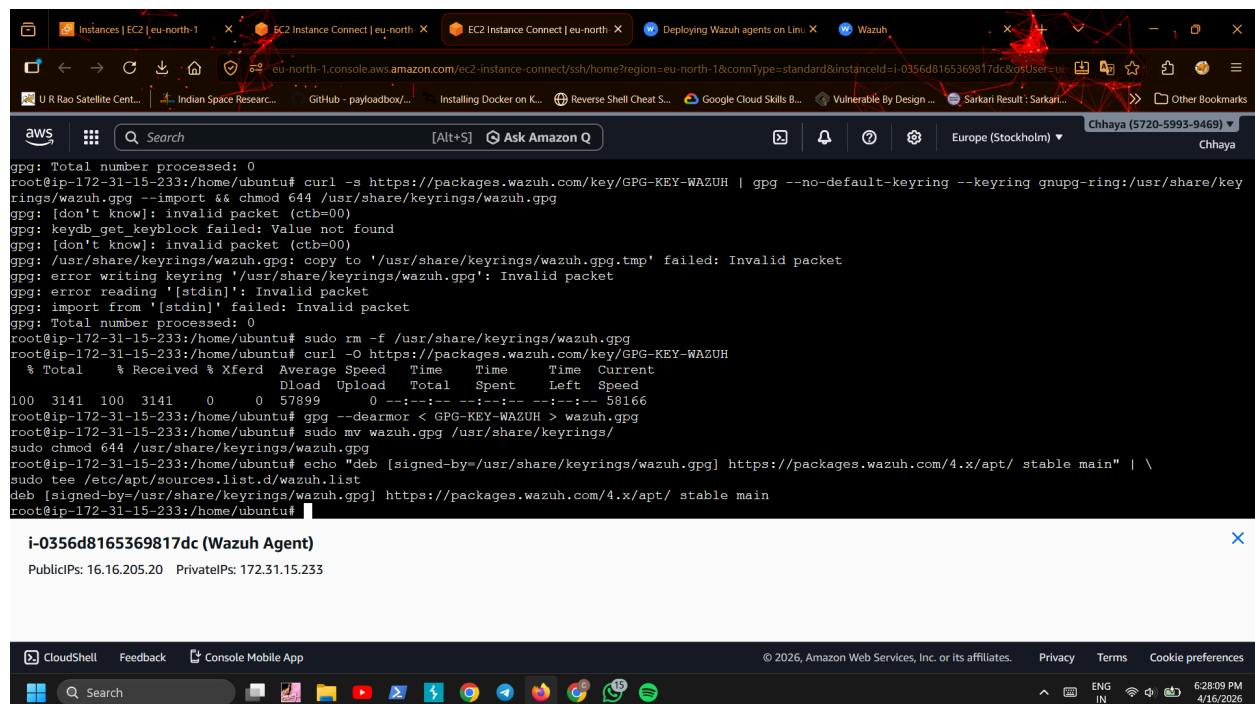
The Wazuh Agent is responsible for collecting logs and system activity from the endpoint and sending them securely to the Wazuh Server for analysis.

6.1 Adding Wazuh Repository

First, I added the official Wazuh repository on the Wazuh Agent machine.

- Installed required packages for repository handling
- Added the Wazuh GPG key for secure package verification
- Added Wazuh repository to the system
- Updated package list

This step ensures the system can download the official Wazuh agent package.



```
gpg: Total number processed: 0
root@ip-172-31-15-233:/home/ubuntu# curl -s https://packages.wazuh.com/key/GPG-KEY-WAZUH | gpg --no-default-keyring --keyring gnupg-ring:/usr/share/keyrings/wazuh.gpg --import && chmod 644 /usr/share/keyrings/wazuh.gpg
gpg: [don't know]: invalid packet (ctb=00)
gpg: keydb get keyblock failed: Value not found
gpg: [don't know]: invalid packet (ctb=00)
gpg: /usr/share/keyrings/wazuh.gpg: copy to '/usr/share/keyrings/wazuh.gpg.tmp' failed: Invalid packet
gpg: error writing keyring '/usr/share/keyrings/wazuh.gpg': Invalid packet
gpg: error reading '[stdin]': Invalid packet
gpg: import from '[stdin]' failed: Invalid packet
gpg: Total number processed: 0
root@ip-172-31-15-233:/home/ubuntu# sudo rm -f /usr/share/keyrings/wazuh.gpg
root@ip-172-31-15-233:/home/ubuntu# curl -O https://packages.wazuh.com/key/GPG-KEY-WAZUH
% Total % Received % Xferd Average Speed Time Time Time Current
Dload Upload Total Spent Left Speed
100 3141 100 3141 0 0 57899 0 --:--:-- --:--:-- --:--:-- 58166
root@ip-172-31-15-233:/home/ubuntu# gpg --dearmor < GPG-KEY-WAZUH > wazuh.gpg
root@ip-172-31-15-233:/home/ubuntu# sudo mv wazuh.gpg /usr/share/keyrings/
root@ip-172-31-15-233:/home/ubuntu# sudo chmod 644 /usr/share/keyrings/wazuh.gpg
root@ip-172-31-15-233:/home/ubuntu# echo "deb [signed-by=/usr/share/keyrings/wazuh.gpg] https://packages.wazuh.com/4.x/apt/ stable main" | \
sudo tee /etc/apt/sources.list.d/wazuh.list
deb [signed-by=/usr/share/keyrings/wazuh.gpg] https://packages.wazuh.com/4.x/apt/ stable main
root@ip-172-31-15-233:/home/ubuntu#
```

i-0356d8165369817dc (Wazuh Agent)

PublicIPs: 16.16.205.20 PrivateIPs: 172.31.15.233

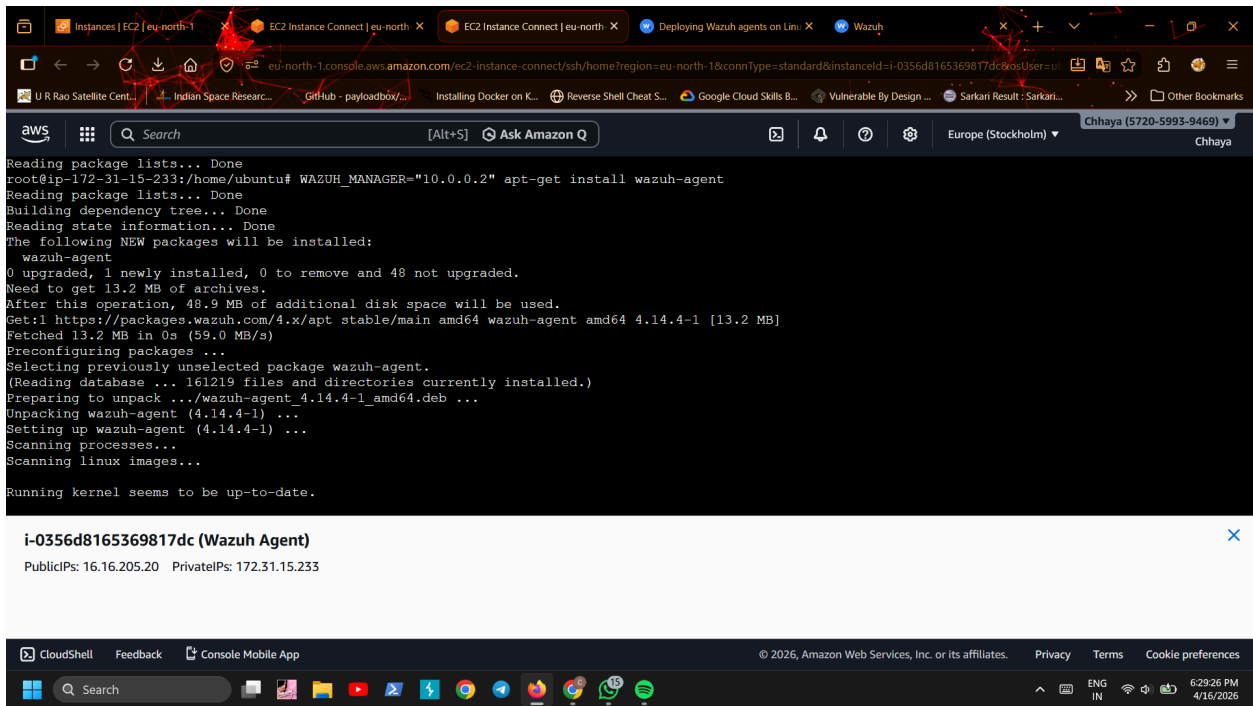
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Search ENG IN 6:28:09 PM 4/16/2026

6.2 Installing Wazuh Agent

After adding the repository, I installed the Wazuh agent.

- Installed the Wazuh agent package on the system
- While installing, I provided the **Wazuh Server IP address** so that the agent can automatically connect to the server
- This step links the agent with the central SIEM system



```
Reading package lists... Done
root@ip-172-31-15-233:/home/ubuntu# WAZUH_MANAGER="10.0.0.2" apt-get install wazuh-agent
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following NEW packages will be installed:
  wazuh-agent
0 upgraded, 1 newly installed, 0 to remove and 48 not upgraded.
Need to get 13.2 MB of archives.
After this operation, 48.9 MB of additional disk space will be used.
Get:1 https://packages.wazuh.com/4.x/apt stable/main amd64 wazuh-agent amd64 4.14.4-1 [13.2 MB]
Fetched 13.2 MB in 0s (59.0 MB/s)
Preconfiguring packages ...
Selecting previously unselected package wazuh-agent.
(Reading database ... 161219 files and directories currently installed.)
Preparing to unpack ../wazuh-agent_4.14.4-1_amd64.deb ...
Unpacking wazuh-agent (4.14.4-1) ...
Setting up wazuh-agent (4.14.4-1) ...
Scanning processes...
Scanning linux images...

Running kernel seems to be up-to-date.
```

i-0356d8165369817dc (Wazuh Agent)

PublicIPs: 16.16.205.20 PrivateIPs: 172.31.15.233

6.3 Starting Wazuh Agent Service

After installation:

- Reloaded system services
- Enabled Wazuh agent to start automatically on boot
- Started the Wazuh agent service

Once started, the agent began communicating with the Wazuh Server.

No user sessions are running outdated binaries.

No VM guests are running outdated hypervisor (qemu) binaries on this host.

```
root@ip-172-31-15-233:/home/ubuntu# systemctl daemon-reload
systemctl enable wazuh-agent
systemctl start wazuh-agent
Created symlink /etc/systemd/system/multi-user.target.wants/wazuh-agent.service → /usr/lib/systemd/system/wazuh-agent.service.
root@ip-172-31-15-233:/home/ubuntu# sed -i "s/^deb/#deb/" /etc/apt/sources.list.d/wazuh.list
apt-get update
Hit:1 http://eu-north-1.ec2.archive.ubuntu.com/ubuntu noble InRelease
Hit:2 http://eu-north-1.ec2.archive.ubuntu.com/ubuntu noble-updates InRelease
Hit:3 http://eu-north-1.ec2.archive.ubuntu.com/ubuntu noble-backports InRelease
Ign:4 http://security.ubuntu.com/ubuntu noble-security InRelease
Get:4 http://security.ubuntu.com/ubuntu noble-security InRelease [126 kB]
Get:5 http://security.ubuntu.com/ubuntu noble-security/main amd64 Packages [1608 kB]
Get:6 http://security.ubuntu.com/ubuntu noble-security/main Translation-en [257 kB]
Ign:7 http://security.ubuntu.com/ubuntu noble-security/restricted amd64 Packages
Get:8 http://security.ubuntu.com/ubuntu noble-security/restricted Translation-en [658 kB]
Ign:7 http://security.ubuntu.com/ubuntu noble-security/restricted amd64 Packages
Get:7 http://security.ubuntu.com/ubuntu noble-security/restricted amd64 Packages [2822 kB]
Fetched 5472 kB in 3min 27s (26.4 kB/s)
Reading package lists... Done
root@ip-172-31-15-233:/home/ubuntu#
```

i-0356d8165369817dc (Wazuh Agent)

PublicIPs: 16.16.205.20 PrivateIPs: 172.31.15.233

CloudShell Feedback Console Mobile App

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Search

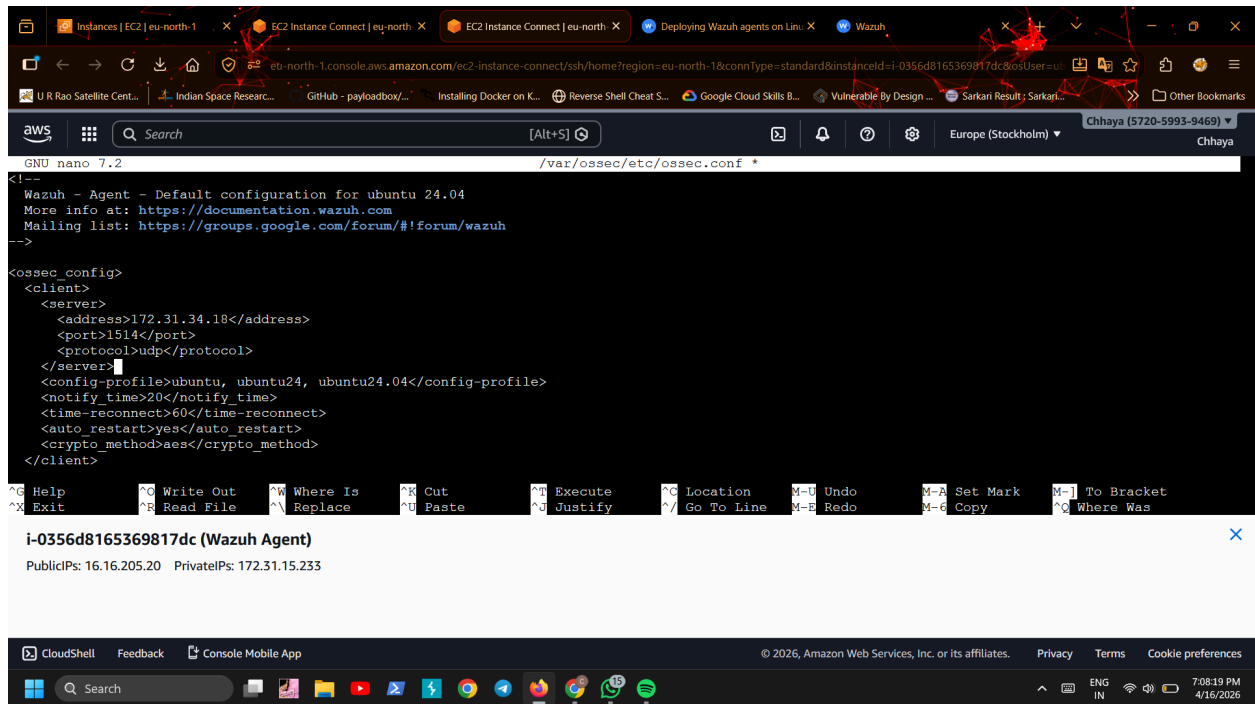
ENG IN 6:43:31 PM 4/16/2026

6.4 Verifying Agent Connection

After starting the service:

- Checked the Wazuh dashboard on the **Wazuh Server**
- Verified that the **Wazuh Agent** was listed as active
- Confirmed successful communication between agent and server

This confirmed that log forwarding was working correctly.



```
GNU nano 7.2 /var/ossec/etc/ossec.conf
Wazuh - Agent - Default configuration for ubuntu 24.04
More info at: https://documentation.wazuh.com
Mailing list: https://groups.google.com/forum/#!forum/wazuh
-->
<ossec_config>
<client>
  <server>
    <address>172.31.34.18</address>
    <port>1514</port>
    <protocol>udp</protocol>
  </server>
  <config-profile>ubuntu, ubuntu24, ubuntu24.04</config-profile>
  <notify_time>20</notify_time>
  <time-reconnect>60</time-reconnect>
  <auto_restart>yes</auto_restart>
  <crypto_method>aes</crypto_method>
</client>
```

i-0356d8165369817dc (Wazuh Agent)
PublicIPs: 16.16.205.20 PrivateIPs: 172.31.15.233

6.5 Disabling Repository Updates (Optional Step)

After successful installation, I disabled automatic updates for Wazuh packages to avoid accidental version mismatches.

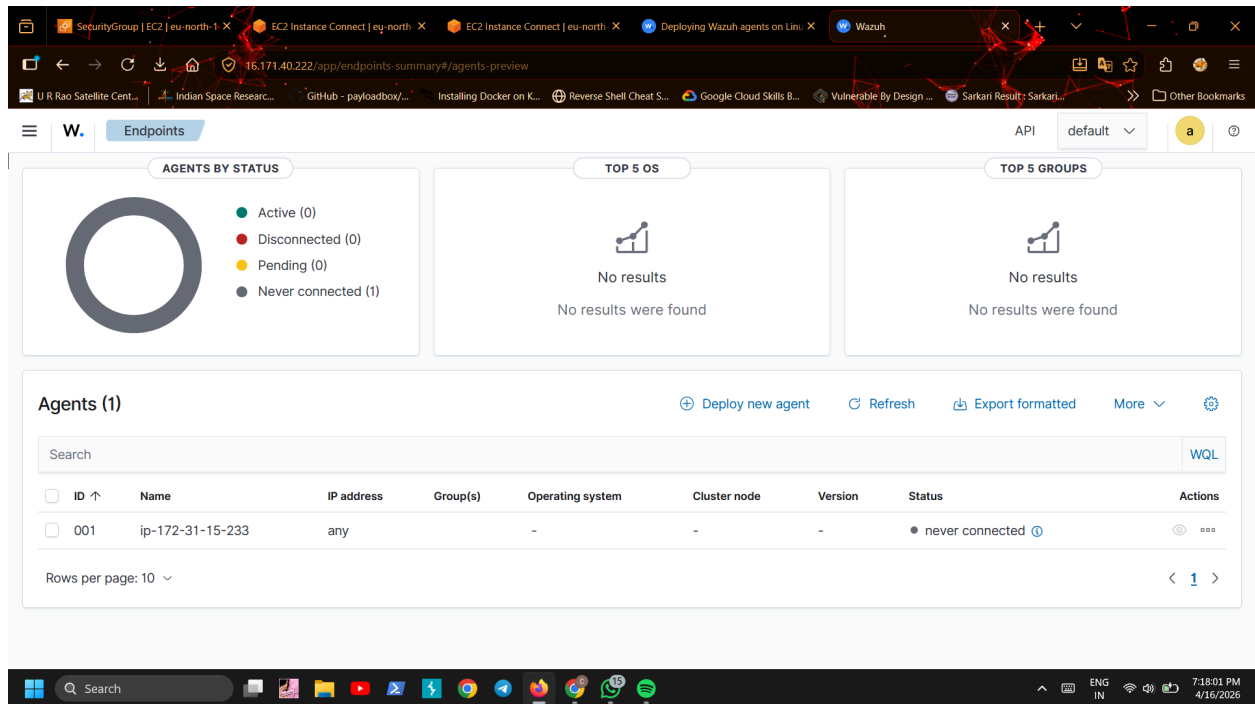
This ensures compatibility between the Wazuh Agent and Wazuh Server.

6.6 Final Status Check

At the end of the setup:

- Wazuh Agent was installed successfully
- It was connected to the Wazuh Server
- Logs from the agent were being received in the dashboard

This confirmed that the endpoint monitoring setup was working correctly.



Detection & Prevention Lab

1. Objective

- * Detect network scans (e.g., Nmap)
- * Log firewall activity
- * Send logs to Wazuh SIEM
- * Trigger alerts
- * (Optional) Block attacker automatically

2. Architecture

text

Attacker → Agent (UFW/iptables logs)

- Wazuh Agent collects logs
- Wazuh Manager analyzes logs
- Alerts generated in Dashboard

→ Active response blocks attacker (optional)

3. Environment

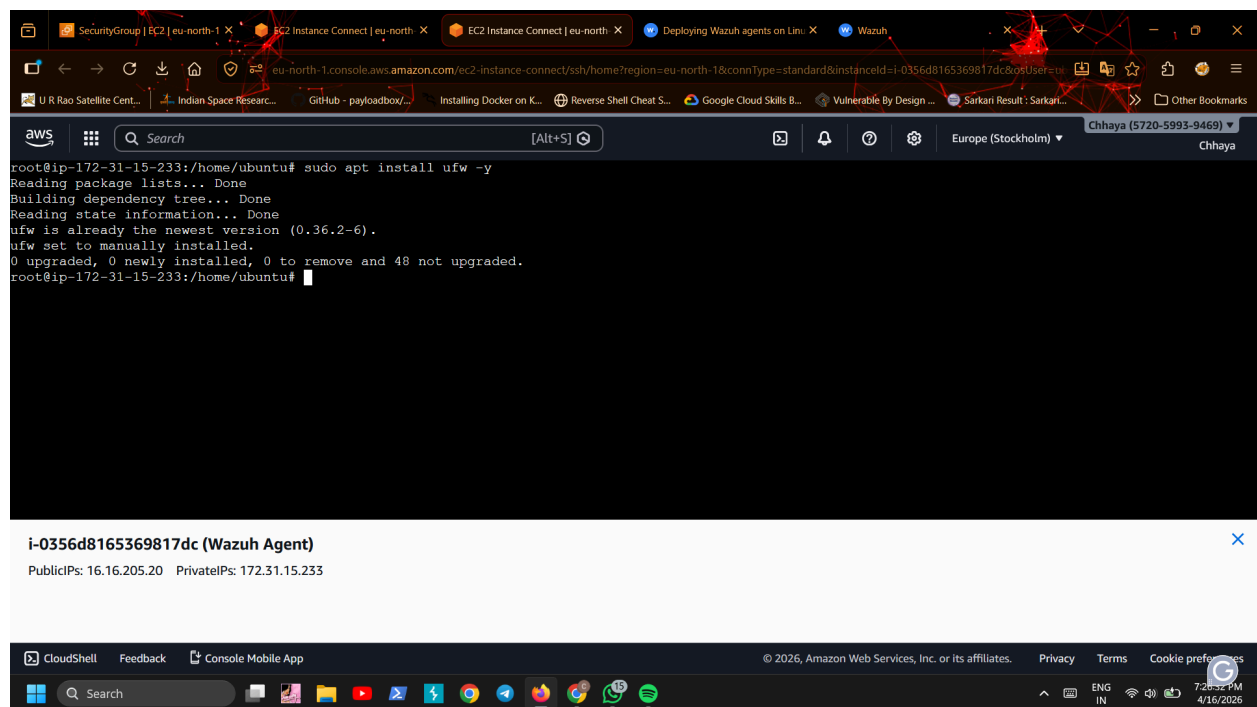
Component	IP	
Wazuh Manager	172.31.34.225	
Wazuh Agent	172.31.2.238	
OS	Ubuntu	
Firewall	UFW + iptables	

4. Firewall Configuration (Agent)

4.1 Install UFW

bash

```
sudo apt install ufw -y
```



4.2 Allow Required Ports

bash

```
sudo ufw allow 22/tcp    # SSH (IMPORTANT)
```

```
sudo ufw allow 1514/udp    # Wazuh logs
```

```
sudo ufw allow 1515/tcp    # Agent registration
```

```
sudo ufw allow from 172.31.34.225 # Allow manager
```

4.3 Enable Logging

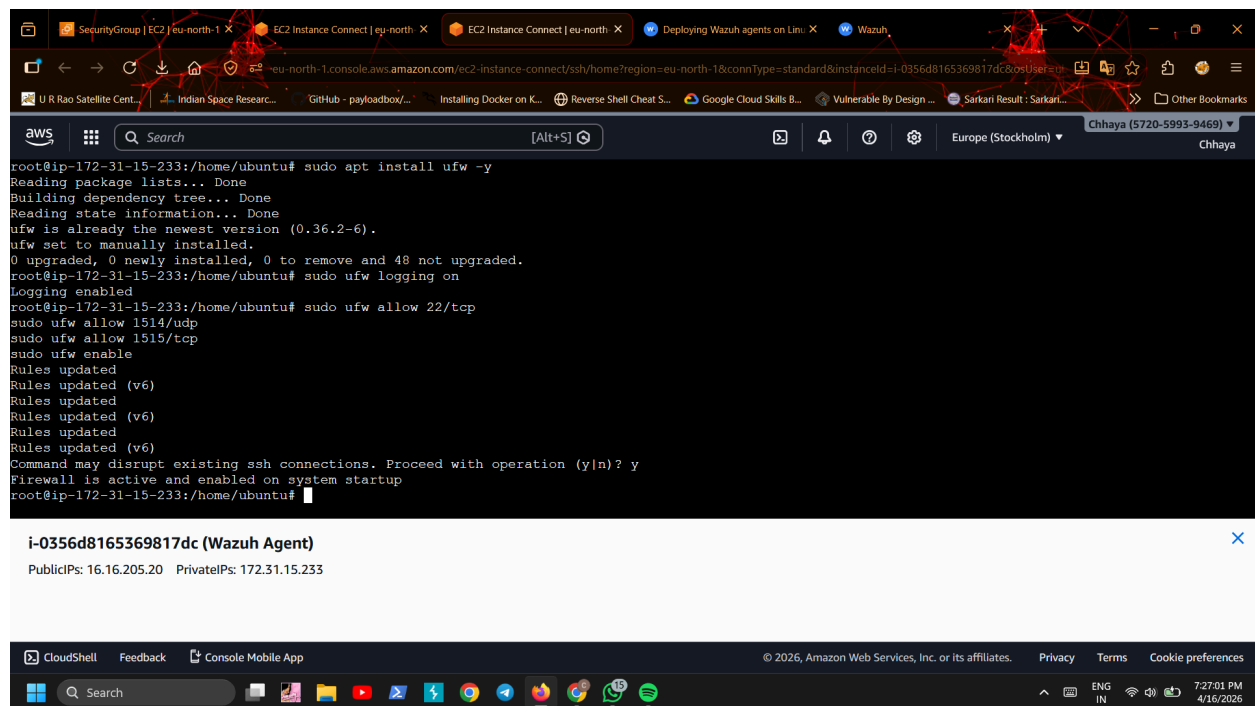
```
bash
```

```
sudo ufw logging on
```

4.4 Enable Firewall

```
bash
```

```
sudo ufw enable
```



```
root@ip-172-31-15-233:/home/ubuntu# sudo apt install ufw -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
ufw is already the newest version (0.36.2-6).
ufw set to manually installed.
0 upgraded, 0 newly installed, 0 to remove and 48 not upgraded.
root@ip-172-31-15-233:/home/ubuntu# sudo ufw logging on
Logging enabled
root@ip-172-31-15-233:/home/ubuntu# sudo ufw allow 22/tcp
sudo ufw allow 1514/udp
sudo ufw allow 1515/tcp
sudo ufw enable
Rules updated
Rules updated (v6)
Rules updated
Rules updated (v6)
Rules updated
Rules updated (v6)
Command may disrupt existing ssh connections. Proceed with operation (y|n)? y
Firewall is active and enabled on system startup
root@ip-172-31-15-233:/home/ubuntu#
```

i-0356d8165369817dc (Wazuh Agent)

PublicIPs: 16.16.205.20 PrivateIPs: 172.31.15.233

4.5 Verify

```
bash
```

```
sudo ufw status
```

5. Log Generation

Firewall logs stored at:

6.2 Restart agent

```
bash
```

```
systemctl restart wazuh-agent
```

7. Log Flow Verification

* Logs appear in Wazuh Dashboard

* Source: agent (172.31.2.238)

* Program: kernel

* Message contains: UFW BLOCK

8. Custom Rule Creation (Detection)

8.1 Create rule

On Wazuh Manager:

```
bash
```

```
nano /var/ossec/etc/rules/local_rules.xml
```

Rule 1: Basic Detection

```
xml
```

```
<group name="ufw,firewall,attack">
```

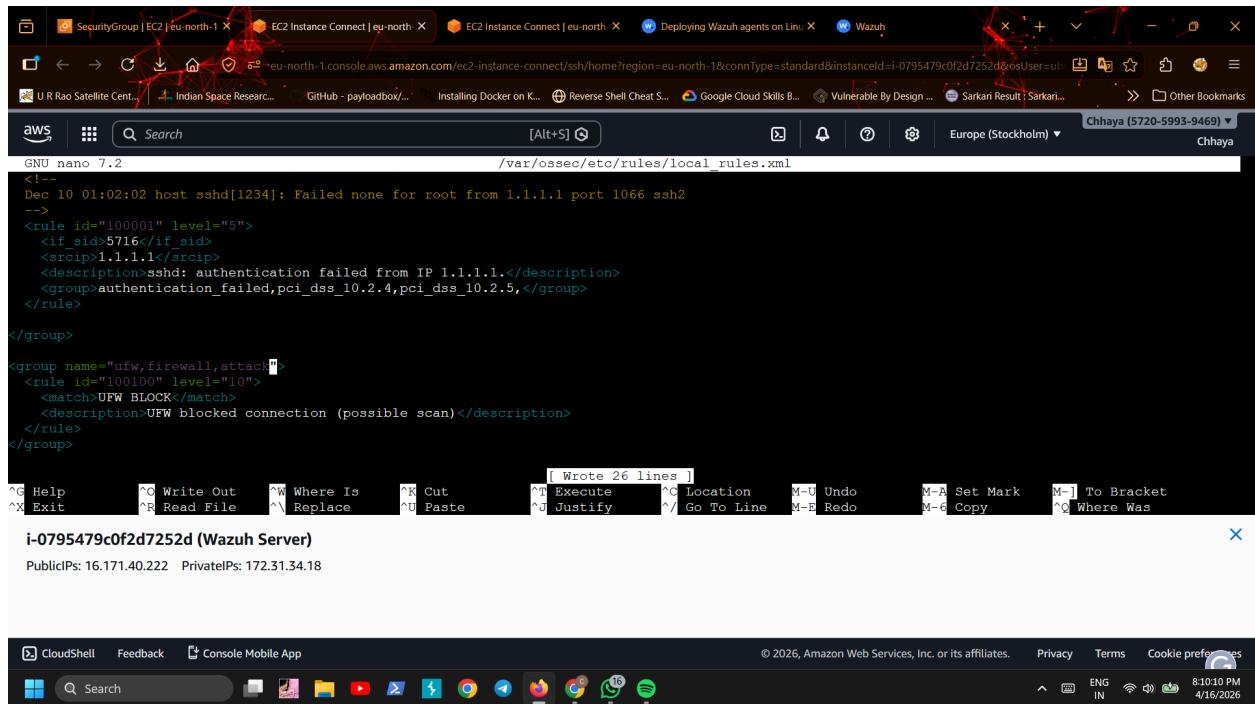
```
  <rule id="100100" level="10">
```

```
    <match>UFW BLOCK </match>
```

```
    <description>UFW blocked connection (possible scan) </description>
```

```
  </rule>
```

```
</group>
```



8.2 Restart manager

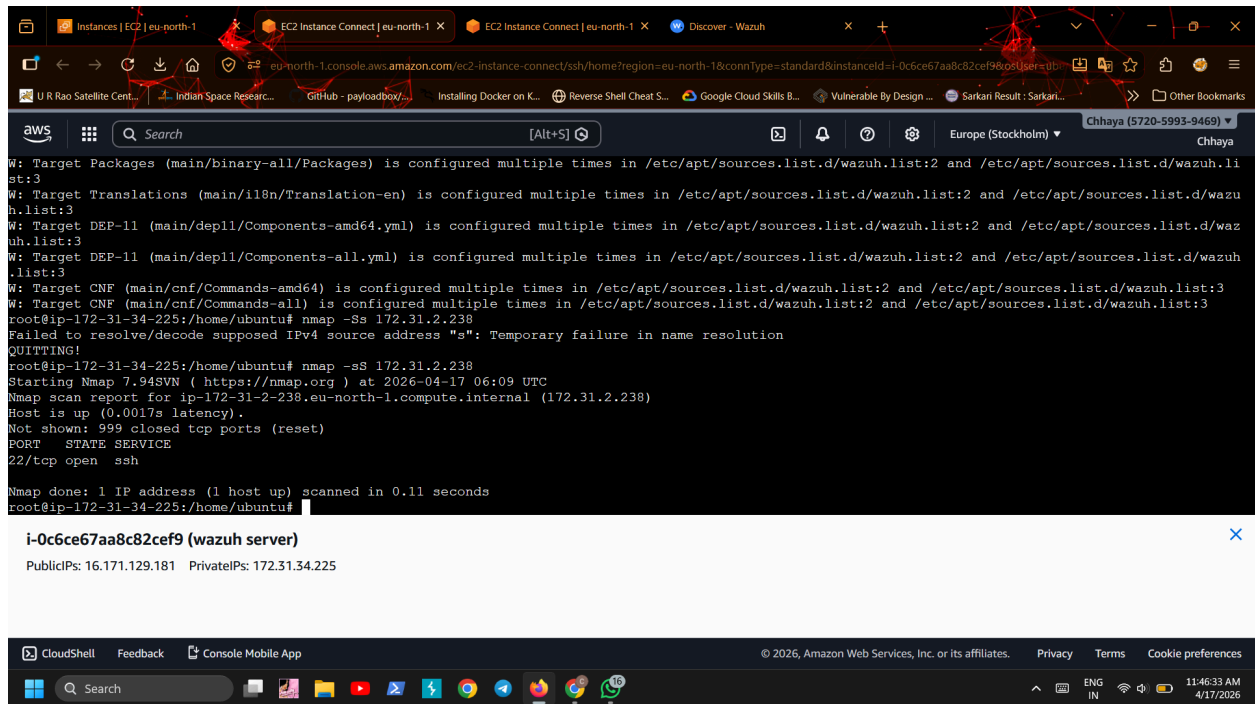
bash

systemctl restart wazuh-manager

9. Testing

From attacker machine:

nmap -sS 172.31.2.238



10. Expected Results

Alert 1:

* Rule ID: 100100

* Description: UFW blocked connection

* Level: 10

11. Troubleshooting Steps

Issue 1: Agent not visible

Fixed by:

bash

agent-auth -m <manager_ip>

Issue 2: Port 1515 blocked

Fixed by:

* AWS Security Group → allow TCP 1515

Issue 3: No UFW logs in Wazuh

Fixed by:

* Adding /var/log/ufw.log in ossec.conf

Issue 4: Rule not triggering

Cause:

* Wrong SID or incorrect rule

Fix:

xml

```
<match>UFW BLOCK</match>
```

Issue 5: Manager blocked by firewall

Fixed by:

bash

ufw allow from 172.31.2.238

The screenshot shows an AWS CloudShell terminal window. The top bar displays the AWS logo, a search bar, and the user 'Chhaya (5720-5993-9469)'. The terminal output shows a series of network logs from the kernel, including UFW ALLOW, UFW BLOCK, and UFW AUDIT messages. The logs indicate traffic from 172.31.2.238 to various destinations. Below the logs, a summary box for the public IP 16.16.99.187 is shown, along with the private IP 172.31.2.238. The bottom of the terminal shows the CloudShell interface with a search bar and system icons.

```
2026-04-17T06:16:36.631193+00:00 ip-172-31-2-238 kernel: [UFW ALLOW] IN= OUT=ens5 SRC=172.31.2.238 DST=185.125.190.56 LEN=76 TOS=0x00 PREC=0x00 TTL=64 ID=31913 DF PROTO=UDP SPT=45318 DPT=123 LEN=56
2026-04-17T06:16:36.831202+00:00 ip-172-31-2-238 kernel: [UFW ALLOW] IN= OUT=ens5 SRC=172.31.2.238 DST=91.189.91.157 LEN=76 TOS=0x00 PREC=0x00 TTL=64 ID=3673 DF PROTO=UDP SPT=40350 DPT=123 LEN=56
2026-04-17T06:16:37.629203+00:00 ip-172-31-2-238 kernel: [UFW ALLOW] IN= OUT=ens5 SRC=172.31.2.238 DST=169.254.169.123 LEN=76 TOS=0x00 PREC=0x00 TTL=64 ID=19063 DF PROTO=UDP SPT=47967 DPT=123 LEN=56
2026-04-17T06:16:49.572560+00:00 ip-172-31-2-238 kernel: [UFW AUDIT] IN=ens5 OUT= MAC=0e:2e:75:97:c3:dd:0e:0b:86:35:00:59:08:00 SRC=147.185.132.168 DST=172.31.2.238 LEN=44 TOS=0x00 PREC=0x00 TTL=248 ID=54321 PROTO=TCP SPT=57342 DPT=7443 WINDOW=65535 RES=0x00 SYN URG=0
2026-04-17T06:16:49.572582+00:00 ip-172-31-2-238 kernel: [UFW BLOCK] IN=ens5 OUT= MAC=0e:2e:75:97:c3:dd:0e:0b:86:35:00:59:08:00 SRC=147.185.132.168 DST=172.31.2.238 LEN=44 TOS=0x00 PREC=0x00 TTL=248 ID=54321 PROTO=TCP SPT=57342 DPT=7443 WINDOW=65535 RES=0x00 SYN URG=0
2026-04-17T06:16:49.573217+00:00 ip-172-31-2-238 kernel: [UFW AUDIT] IN= OUT=ens5 SRC=172.31.2.238 DST=13.48.4.203 LEN=632 TOS=0x10 PREC=0x00 TTL=64 ID=24919 DF PROTO=TCP SPT=22 DPT=49698 WINDOW=249 RES=0x00 ACK PSH URG=0
2026-04-17T06:16:49.928082+00:00 ip-172-31-2-238 kernel: [UFW BLOCK] IN=ens5 OUT= MAC=0e:2e:75:97:c3:dd:0e:0b:86:35:00:59:08:00 SRC=147.185.132.134 DST=172.31.2.238 LEN=44 TOS=0x00 PREC=0x00 TTL=248 ID=54321 PROTO=TCP SPT=49690 DPT=1719 WINDOW=65535 RES=0x00 SYN URG=0
2026-04-17T06:16:52.414506+00:00 ip-172-31-2-238 kernel: [UFW BLOCK] IN=ens5 OUT= MAC=0e:2e:75:97:c3:dd:0e:0b:86:35:00:59:08:00 SRC=64.62.156.31 DST=172.31.2.238 LEN=40 TOS=0x00 PREC=0x00 TTL=242 ID=54321 PROTO=TCP SPT=51826 DPT=8123 WINDOW=65535 RES=0x00 SYN URG=0
2026-04-17T06:16:53.808188+00:00 ip-172-31-2-238 kernel: [UFW ALLOW] IN= OUT=ens5 SRC=172.31.2.238 DST=169.254.169.123 LEN=76 TOS=0x00 PREC=0x00 TTL=64 ID=63389 DF PROTO=UDP SPT=52588 DPT=123 LEN=56
2026-04-17T06:16:55.068998+00:00 ip-172-31-2-238 kernel: [UFW BLOCK] IN=ens5 OUT= MAC=0e:2e:75:97:c3:dd:0e:0b:86:35:00:59:08:00 SRC=198.235.24.207 DST=172.31.2.238 LEN=44 TOS=0x00 PREC=0x00 TTL=249 ID=61163 PROTO=TCP SPT=52624 DPT=5904 WINDOW=1024 RES=0x00 SYN URG=0
2026-04-17T06:16:56.589097+00:00 ip-172-31-2-238 kernel: [UFW BLOCK] IN=ens5 OUT= MAC=0e:2e:75:97:c3:dd:0e:0b:86:35:00:59:08:00 SRC=100.29.192.98 DST=172.31.2.238 LEN=44 TOS=0x00 PREC=0x00 TTL=244 ID=54321 PROTO=TCP SPT=6680 DPT=22222 WINDOW=65535 RES=0x00 SYN URG=0
```

i-0cdc0fa7194277c57 (wazuh agent)

Public IPs: 16.16.99.187 Private IPs: 172.31.2.238

Issue 6: Dashboard “Server not ready”

Fixed by:

* Correct wazuh.yml config

* API authentication

* Clearing cache

12. Key Learnings

* UFW generates firewall logs

* Wazuh parses and correlates logs

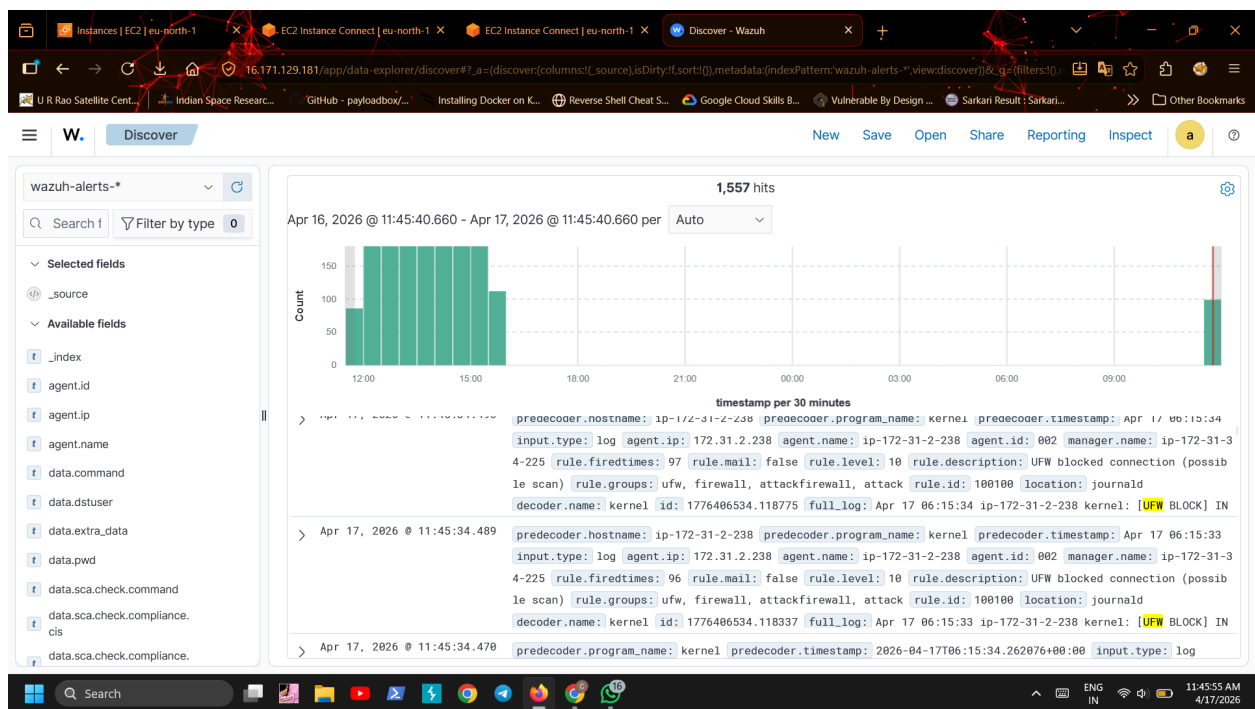
* Custom rules detect threats

* Frequency-based rules detect scans

* Active response enables auto-blocking

13. Final Outcome

- Real-time firewall monitoring
- Detection of external scans
- Alert generation in SIEM
- Capability to auto-block attackers



14. Alerting Configuration (Wazuh Dashboard / OpenSearch)

14.1 Navigate to Alerting

In Wazuh Dashboard:

Go to:

Menu → OpenSearch Plugins → Alerting → Monitors

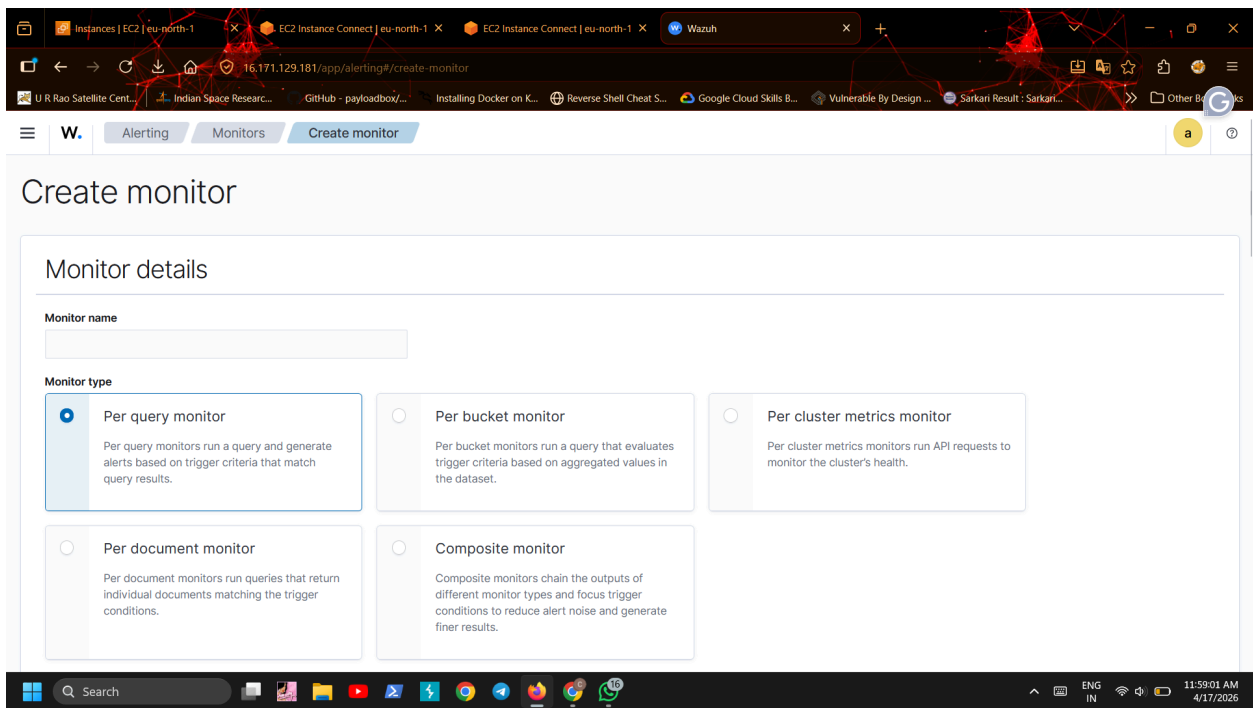
14.2 Create Monitor

Click: *Create monitor*

Select:

text

Monitor type: Per query monitor



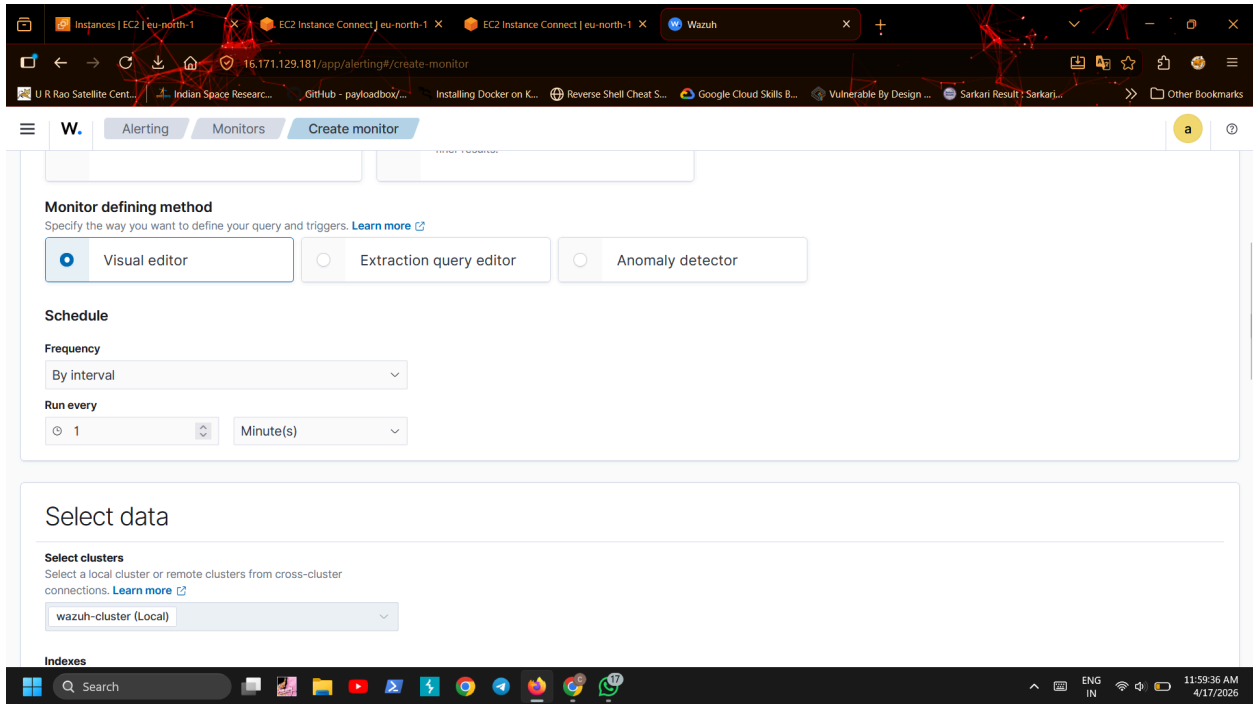
Basic details

| Field | Value |

| Name | UFW Scan Detection Monitor |

| Index | wazuh-alerts-* |

| Schedule | Every 1 minute |



What this does

* Matches your custom rules:

* 100100 → single block

* 100101 → scan detected

14.3 Define Trigger

Click:

Add trigger

Fill:

Field	Value
-----	-----
Trigger name	UFW Attack Detected
Severity	High

Trigger condition

painless

ctx.results[0].hits.total.value > 0

Meaning

If ANY matching alert exists → trigger fires

Create

The screenshot displays the Wazuh alerting interface in a web browser. The browser's address bar shows the URL: `15.174.129.181/app/alerting#/monitors/7znYkp0Bc8mEJfFRxLZI/alertState=ALL&from=0&monitorIds=7znYkp0Bc8mEJfFRxLZI&monitorType=query_level_monitor&`. The browser's tab bar includes several tabs: 'Instances | EC2 | eu-north-1', 'EC2 Instance Connect | eu-north-1', 'EC2 Instance Connect | eu-north-1', and 'Wazuh'. The Wazuh interface has a navigation bar with 'Alerting', 'Monitors', and 'Nmap scan' tabs. The 'Nmap scan' monitor is currently selected and is in an 'Enabled' state. The interface includes buttons for 'Edit', 'Disable', 'Export as JSON', and 'Delete'. The 'Overview' section provides details about the monitor: 'Monitor type' is 'Per query monitor', 'Monitor definition type' is 'Visual Graph', 'Index' is 'wazuh-alerts-*' (with a 'View all 2' link), 'Total active alerts' is '1', 'Schedule' is 'Every 1 minutes', 'Last updated' is '04/16/26 2:02 am IST', 'Monitor ID' is '7znYkp0Bc8mEJfFRxLZI', and 'Monitor version number' is '1'. The 'Associations with composite monitors' section is empty. The 'Triggers (1)' section shows a table with one trigger: 'Nmap scan alert', which has '0' actions and a 'Severity' of '1'. The bottom of the screenshot shows a Windows taskbar with various application icons and a system tray indicating the time as 12:05:14 PM on 4/17/2026.

Nmap scan • Enabled

Overview

Monitor type Per query monitor	Monitor definition type Visual Graph	Index wazuh-alerts-* View all 2	Total active alerts 1
Schedule Every 1 minutes	Last updated 04/16/26 2:02 am IST	Monitor ID 7znYkp0Bc8mEJfFRxLZI	Monitor version number 1
Associations with composite monitors -			

Triggers (1)

Name ↑	Number of actions	Severity
Nmap scan alert	0	1

15. Testing Alerting

Step 1 — Trigger attack

From attacker machine:

```
bash
```

```
nmap -sS 172.31.2.238
```

W: Target Packages (main/binary-all/Packages) is configured multiple times in /etc/apt/sources.list.d/wazuh.list:2 and /etc/apt/sources.list.d/wazuh.list:3
W: Target Translations (main/il8n/Translation-en) is configured multiple times in /etc/apt/sources.list.d/wazuh.list:2 and /etc/apt/sources.list.d/wazuh.list:3
W: Target DEP-11 (main/depl1/Components-amd64.yml) is configured multiple times in /etc/apt/sources.list.d/wazuh.list:2 and /etc/apt/sources.list.d/wazuh.list:3
W: Target DEP-11 (main/depl1/Components-all.yml) is configured multiple times in /etc/apt/sources.list.d/wazuh.list:2 and /etc/apt/sources.list.d/wazuh.list:3
W: Target CNF (main/cnf/Commands-amd64) is configured multiple times in /etc/apt/sources.list.d/wazuh.list:2 and /etc/apt/sources.list.d/wazuh.list:3
W: Target CNF (main/cnf/Commands-all) is configured multiple times in /etc/apt/sources.list.d/wazuh.list:2 and /etc/apt/sources.list.d/wazuh.list:3
root@ip-172-31-34-225:/home/ubuntu# nmap -sS 172.31.2.238
Failed to resolve/decode supposed IPv4 source address "s": Temporary failure in name resolution
QUITTING!
root@ip-172-31-34-225:/home/ubuntu# nmap -sS 172.31.2.238
Starting Nmap 7.94SVN (https://nmap.org) at 2026-04-17 06:09 UTC
Nmap scan report for ip-172-31-2-238.eu-north-1.compute.internal (172.31.2.238)
Host is up (0.0017s latency).
Not shown: 999 closed tcp ports (reset)
PORT STATE SERVICE
22/tcp open ssh
Nmap done: 1 IP address (1 host up) scanned in 0.11 seconds
root@ip-172-31-34-225:/home/ubuntu#

i-0c6ce67aa8c82cef9 (wazuh server)

PublicIPs: 16.171.129.181 PrivateIPs: 172.31.34.225

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Step 2 — Verify

Go to:

Alerting → Alerts

16.171.129.181/app/alerting#/dashboard?alertState=ALL&from=0&search=&severityLevel=ALL&size=10000&sortDirection=desc&sortField=start_time

W. Alerting Alerts

Alerts Monitors Destinations

Alerts by triggers

Search

All severity levels All alerts

☐ Alerts	Active	Acknowledged	Errors	Trigger name	Trigger start time ↓	Trigger last updated	Severity	Monitor name
☐ 3 alerts	1	0	0	Nmap scan alert	04/17/26 11:34 am	04/17/26 12:06 pm	1	Nmap scan

< 1 >

Expected result

- * Monitor triggered
- * Alert created
- * Severity: High

16. Sample Alert Output

Monitor: UFW Scan Detection Monitor

Trigger: UFW Attack Detected

Rule: Port scan detected (multiple UFW blocks)

Source IP: 147.185.x.x

17. Final Workflow

Attacker →

UFW blocks →

Wazuh detects →

Custom rule fires →

Monitor detects →

Trigger fires →

Alert generated →

Notification sent

Final Summary

A monitoring rule was created in the Wazuh dashboard to detect firewall-based alerts generated from custom rules. When suspicious activity such as port scanning is identified, the system triggers alerts and optionally sends notifications for incident response.

